① (B) $x^2y' + 2xy = y^3$

$y' + p(x)y = q(x)y^n$

This makes it a Bernoulli's equ

$\Rightarrow y' + \frac{2}{x}y = y^3x^{-2}$

Dividing thru by y^3

$y^{-3}y' + \frac{2}{x}y^{-2} = x^{-2}$

Let $v = y^{-2}$

② $M = (xy^2 + bx^2y)dx$

$N = (x+y)x^2dy$

$M_y = 2xy + bx^2$

$N_x = 3x^2 + 2xy$

For exactness $M_y = N_x$

$2xy + bx^2 = 3x^2 + 2xy$

$b = 3$ C

③ $y' - 2y = x^2e^{2x}$

$y' + p(x)y = q(x)$

$p(x) = -2$

$\mu = e^{\int p(x)dx}$

$\mu = e^{-\int 2\,dx}$

$\mu = e^{-2x}$

$ye^{-2x} = \int x^2$

$ye^{-2x} = \frac{x^3}{3} + c$

$y = \frac{x^3e^{2x}}{3} + ce^{2x}$

$y = \left(\frac{x^3}{3} + c\right)e^{2x}$

D

1

④ $y' = -2 - y + y^2$ — (*) Ricc

where $y_1 = 2$

$y = \frac{1}{z} + 2$ — ① $\quad \frac{dy}{dx} = \frac{-1}{z^2}\frac{dz}{dx}$ — ②

Substituting ① and ② into eqn (*)

$$\frac{-1}{z^2}\frac{dz}{dx} = -2 - \frac{1}{z} - 2 + \left(\frac{1}{z} + 2\right)$$

$$= -4 - \frac{1}{z} + \frac{1}{z^2} + \frac{4}{z} + 4$$

$$\frac{-1}{z^2}\frac{dz}{dx} = \frac{1}{z^2} + \frac{3}{z}$$

$$\frac{dz}{dx} = -1 + 3z$$

$z' - 3z = -1$

$y' + p(x)y = q(x)$

$p(x) = -3$

$\mu = e^{\int p(x)\,dx}$

$\mu = e^{\int 3\,dx}$

$\mu = e^{3x}$

$ze^{3x} = \int -e^{3x}$

$\Rightarrow ze^{3x} = \frac{-1}{3}e^{3x} + C$

$z = \frac{-1}{3} + Ce^{-3x}$

From eqn ①

$z = \frac{1}{y-2}$

$\frac{1}{y-2} = \frac{-1}{3} + Ce^{-3x}$

$y - 2 = \frac{1}{-\frac{1}{3} + Ce^{-3x}}$

$y = 2 + \frac{1}{-\frac{1}{3} + ce^{-3x}} \Rightarrow y = 2 - \frac{1}{\frac{1}{3} + ce^{-3x}}$ C

(5) $y_1(x) = 1 \quad y_2(x) = e^x \quad y_3(x) = e^{-x}$

$$W = \begin{vmatrix} 1 & e^x & e^{-x} \\ 0 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix}$$

Det $1(e^x + e^x) - e^x(e^x - 0) + e^x(e^x - 0)$

at $x = 0$

$1(e^0 + e^0) - e^0(e^0 - 0) + e^0(e^0 - 0)$

$1(2) - 1 + 1 = 2$ D

(7) $2y'' + 2y' - 12y = 0$

GS $= Ge^{2x} + Ge^{-3x}$ A

(6) $y'' + 2y' - 3y = 0 \quad y(0) = b \quad y'(0) = 3$

$y_c = Ge^x + Ge^{-3x} \quad y' = Ge^x - 3Ge^{-3x}$

$b = G + G_2 \quad 3 = G - 3G_2$

Taking the value of $b = 1$ Since the limit is approaching positive infinity.

$G = \frac{3}{2} \quad G_2 = -\frac{1}{2}$

Substituting G & G_2 in the original eqn and differentiating it proves that it's a solution to the diff. eqn

(8) $y = \sin(Ax + B)$

$\sin^{-1} y = Ax + B$

$\frac{1}{\sqrt{1-y^2}} y' = A$

$y' = A(1-y^2)^{1/2}$ ——— (1)

$y'' = \frac{1}{2} A(-2y)(1-y^2)^{-1/2} \frac{dy}{dx}$

$y'' = -A(1-y^2)^{-1/2} y'y$

$y'' = \frac{-A}{(1-y^2)^{1/2}} y'y \Rightarrow (\sqrt{1-y^2})\, y'' = -Ay'y$ ——— (3)

Making A the subject of (1)

$\frac{y'}{\sqrt{1-y^2}} = A$ ⟶ substitute into (3)

$\Rightarrow (\sqrt{1-y^2})\, y'' = \frac{-y'(y'y)}{\sqrt{1-y^2}}$

$(1-y^2) y'' = -y(y')^2$

$(1-y^2) y'' + y(y')^2 = 0$

B

Solving for y_c

(9) $\pm i, 0$

$y_c = C_1 + C_2 \cos t + C_3 \sin t$

$y_p = A = 1$

$G.S = C_1 + C_2 \cos t + C_3 \sin t + A$

$= A + B\cos t + C\sin t + 1$

B

(10) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ for Eigen values

$\lambda^2 - \text{trace}\lambda + \det = 0$

trace $= 4$ $\det A = 3$

(3) $\lambda^2 - 4\lambda + 3 = 0$

$\lambda_1 = 3$ $\lambda_2 = 1$

Eigen vector for $\lambda_1 = 3$

$(A - \lambda I)V = 0$

$\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$R_1 + R_2 = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$

$v_1 - v_2 = 0$

$v_1 = v_2$

$\Rightarrow v_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Eigen vector for $\lambda = 1$

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$

$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $R_1 - R_2 \rightarrow$

$v_1 + v_2 = 0$

$v_1 = -v_2$ $\begin{pmatrix} -v_2 \\ v_2 \end{pmatrix}$

$\Rightarrow v_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$G.S = C_1 v_1 e^{\lambda_1 t} + C_2 v_2 e^{\lambda_2 t}$

$= C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} + C_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{t}$

$= C_1 e^{3t} + C_2 e^{t}$

$= C_1 e^{3t} + C_2 e^{t}$

At $x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$1 = C_1 - C_2$ —— (1)

$1 = C_1 + C_2$ —— (2)

$C_1 = 1$ $C_2 = 0$

$x(t) = \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix}$

(12a) $y' = \frac{y^2-1}{x}$

$\frac{dy}{dx} = \frac{y^2-1}{x} \Rightarrow \frac{dy}{y^2-1} = \frac{1}{x}dx$

Separable Hence we integrate both sides.

$\int \frac{dy}{y^2-1} = \int \frac{1}{x}dx$

NB: $\int \frac{1}{x^2-a^2} = \frac{1}{2a}\ln\frac{x-1}{x+1}$

$\frac{1}{2}\ln\frac{y-1}{y+1} = \ln x + \ln c$

$\frac{1}{2}\ln\frac{y-1}{y+1} = \ln xc$

$\ln\left(\frac{y-1}{y+1}\right)^{\frac{1}{2}} = \ln xc$

$\left(\frac{y-1}{y+1}\right)^{\frac{1}{2}} = xc$

$\frac{y-1}{y+1} = x^2c$

NB: Remember, a constant square gives u a constant. Hence we can neglect the square on it.

$y' + \tan(x)y = \cos^2 x$

$p(x) = \tan x$

$\mu = e^{\int p(x)dx}$

$\mu = e^{\int \tan x\,dx}$

$\mu = e^{\ln\sec x}$

$\ln\mu = \ln\sec x \ln e$

$\ln\mu = \ln\sec x$

$\mu = \sec x$

$y\sec x = \int \cos^2 x \cdot \sec x$

$y\sec x = \int \cos x$

$y\sec x = \sin x + C$

$y = \frac{\sin x}{\sec x} + \frac{C}{\sec x}$

$y = \sin x\cos x + C\cos x$

at $y(0) = 2$

$\therefore C = 2$

$y = \sin x\cos x + 2\cos x$

$y = \cos x(\sin x + 2)$

B

$y-1 = yx^2c + x^2c$

$y - yx^2c = x^2c + 1$

$y(1-x^2c) = x^2c+1$

$y = \frac{x^2c+1}{1-x^2c}$

at $y(1) = 2$

$2 = \frac{c+1}{1-c}$

$1-2c = c+1$

$1 = 3c$

$c = \frac{1}{3}$

$y = \frac{x^2\frac{1}{3}+1}{1-\frac{1}{3}x^2}$

$\frac{1-\frac{1}{3}x^2}{\frac{3-x^2}{3}}$

$\times \frac{3}{3-x^2} = \frac{x^2+3}{3-x}$

$y = \frac{x^2+3}{3-x}$ D

(13) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Eigen value $\lambda^2 - \text{trace}\lambda + \det$

trace = 2 det = −3

$\lambda^2 - 2\lambda - 3 = 0$

$\lambda_1 = 3 \quad \lambda_2 = -1$

(14) Eigen vector for $\lambda_1 = 3$

$(A - I\lambda)V = 0$

$\begin{bmatrix} 1-3 & 2 \\ 2 & 1-3 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$

$\Rightarrow \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{matrix} 0 \\ 0 \end{matrix}$

$v_1 - v_2 = 0$

$v_1 = v_2$ $\quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Eigen vector $\lambda = -1$

$\begin{bmatrix} 1-(-1) & 2 \\ 2 & 1-(-1) \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}\begin{matrix} v_1 \\ v_2 \end{matrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$

$v_1 + v_2 = 0$

$v_1 = -v_2$ $\quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$v_1^T = (1, 1)$

$\begin{pmatrix} 1 \\ -1 \end{pmatrix}^T = (1, -1)$ D

the transpose of each gives us the corresponding eigen vectors.

(15) $q.s = C_1 v_1 e^{\lambda_1 t} + C_2 v_2 e^{\lambda_2 t}$

$= A\begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} + B\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$

$= Ae^{3t}(1,1)^T + Be^{-t}(1,-1)$

Ans = E

(16) $\phi(t) = \begin{bmatrix} e^{3t} & e^{-t} \\ e^{3t} & -e^{-t} \end{bmatrix}$

F

(17) $X(0) = (3, 0)$

$x_1(t) = C_1 e^{3t} + C_2 e^{-t}$

$3 = C_1 + C_2$ — (1)

$x_2(t) = C_1 e^{3t} - C_2 e^{-t}$

$0 = C_1 - C_2$ — (2)

Solving simultaneously

$C_1 = \frac{3}{2}$ $C_2 = \frac{3}{2}$

F

(18)

(19)

(18) $\frac{\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y}}{N} = \frac{My - Nx}{N}$

$u = e^{\int g(x)\,dx}$ D

(19) $ye^x = x + C$

$y = xe^{-x} + Ce^{-x}$ — (1)

$y' = e^{-x} - xe^{-x} - Ce^{-x}$

$y' = e^{-x} - (xe^{-x} + Ce^{-x})$

$y' = e^{-x} - y$

$y' + y = e^{-x}$ A

(20) $y = Ae^x + Be^{2x}$

$y_1 = e^x \quad y_2 = e^{2x}$

$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \Rightarrow \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix}$

$(2e^{3x}) - e^x(e^{2x}) = 2e^{3x} - e^{2x}$

$= e^{3x}$ A

(21) $(2x+3y-3)\,dx + (3x+2y-1)\,dy$

Solve both eqns simultaneously

$2x+3y-3=0 \quad \| \quad 3x+2y-1=0$

$2x+3y=3 \quad \| \quad 3x+2y=1$

$x = X + h \qquad y = Y + k$

where (h, k) are solutions of the simultaneous eqns.

$h = 3 \qquad k = -1$

$x = X + 3 \qquad y = Y - 1$

D

(22) $(4x+8y-12) + (8x+16y+20)\frac{dy}{dx}$

The D.E is linear but not homogeneous. Hence can be transformed by

$\frac{a_1}{a_2} = \frac{b_1}{b_2}$

Let $z = 4x + 8y$ — (1)

Substitute (1) into the D.E

$(z-12) + (2z+20)\frac{dy}{dx}$

$\frac{dz}{dx} = 4 + 8\frac{dy}{dx}$ — (2)

$\frac{dz}{dx} = 4 + 8\frac{dy}{dx}$

Rearranging the original eqn gives us → $\frac{-4x-8y+12}{8x+16y+3} = \frac{dy}{dx}$:

↓ Substitute back into eqn(1)

$$z' = 4 + 8\left[\frac{-4x-8y+12}{8x+16y+20}\right]$$

knowing that $z = 4x+8y$

$$z' = 4 + 8\left[\frac{-(z)+12}{2z+20}\right]$$

$$(2z+20)z' = 4(2z+20) + 8(-z+12)$$

$$(2z+20)z' = 8z + 80 - 8z + 96$$

$$(2z+20)\,dz = 176\,dx$$

Seperable. Hence we integrate both sides.

$$\int (2z+20)\,dz = \int 176\,dx$$

$$z^2 + 20z = 176x$$

$$(4x+8y)^2 + 20(4x+8y) = 176x$$

$$(4x+8y)^2 + 80x + 160y = 176x$$

$$(4x+8y)^2 - 96x + 160y = 0$$

A

$u_1' = \tan x \quad u_2' = 1$

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25) $y'' + y = \sec x$

$y_c = A\cos x + B\sin x$

$y_c = Ay_1 + By_2$

$y_1 = \cos x \quad y_2 = \sin$

$W(y_1, y_2) = 1$

Using variation of parameters

$u_1 y_1 + u_2 y_2$

$$u_1 = -\int \frac{y_2(fx)}{W(y_1 y_2)}$$

$$u_1 = -\int \frac{\sin}{\cos} = +\ln\cos x$$

$$u_2 = \int \frac{y_1(fx)}{W(y_1 y_2)} = \int \frac{\cos x}{\cos x}$$

$u_2 = x$

$\therefore u_1 y_1 + u_2 y_2$

$\cos x(\cos x) + x\sin x$

$= C_1 \cos x(\ln|\cos x|) + x\sin x$

B.

(26) $y'' + 4y' + 4y = xe^{2x}$

$y_c = Ce^{2x} + Cxe^{-2x}$

$y_p = (Ax+B)e^{2x}$

Using variation of parameters:

$y_1 = e^{-2x} \quad y_2 = xe^{-2x}$

$y_p = u_1 y_1 + u_2 y_2$

$u_1 = -\int \frac{y_2 f(x)}{W(y_1, y_2)}$

$W(y_1, y_2) \begin{vmatrix} e^{-2x} & xe^{-2x} \\ -2e^{-2x} & e^{-2x} - 2e^{-2x}x \end{vmatrix}$

$e^{-2x}(e^{-2x} - 2xe^{-2x}) + 2e^{-2x}(xe^{-2x})$

$e^{-4x} - 2xe^{-4x} + 2e^{-4x}x = e^{-4x}$

$\Rightarrow u_1 = -\int \frac{xe^{-2x} \cdot xe^{2x}}{e^{-4x}}$

$u_1 = \int x^2 e^{4x}$

Using by parts we have

$u_1 = \frac{-x^2e^{4x}}{4} + \frac{xe^{4x}}{8} - \frac{e^{4x}}{32}$

$u_1 y_1 = e^{-2x}\left(\frac{-x^2e^{4x}}{4} + \frac{xe^{4x}}{8} - \frac{e^{4x}}{32}\right)$

$u_1 y_1 = \frac{-x^2e^{2x}}{4} + \frac{xe^{2x}}{8} - \frac{e^{2x}}{32}$

$u_2 = \int \frac{y_2 f(x)}{W}$

$u_2 = \int \frac{e^{-2x} \cdot xe^{2x}}{e^{-4x}} = \int xe^{4x}$

$u_2 = \frac{xe^{4x}}{4} - \frac{e^{4x}}{16}$

$u_2 y_2 = xe^{-2x} \cdot e^{4x}\left(\frac{x}{4} - \frac{1}{16}\right)$

$= \frac{x^2e^{2x}}{4} - \frac{xe^{2x}}{16}$

$y_p = u_1 y_1 + u_2 y_2$

$= \frac{-x^2e^{2x}}{4} + \frac{xe^{2x}}{8} - \frac{e^{2x}}{32} + \frac{x^2e^{2x}}{4} - \frac{xe^{2x}}{16}$

$$\Rightarrow \frac{x e^{2x}}{16} - \frac{e^{2x}}{32}$$

$$y_p = \left(\frac{2x-1}{32}\right)e^{2x}$$

$$G.S = Ge^{-2x} + C_2 x e^{-2x} + \left(\frac{2x-1}{32}\right)e^{2x}$$

$$= (G + C_2 x)e^{-2x} + \left(\frac{2x-1}{32}\right)e^{2x}$$

D

(27) $y' + y = (xy)^2$

$y' + y = x^2 y^2$ ——

Its a bernoulli's eqxn

~~let y =~~

$y^{-2}y' + y^{-1} = x^2$ —— (*)

let $u = y^{-1}$ —— (1)

$u' = -y^{-2}y'$ —— (2)

Substitute (1) & (2) in (*)

$-u' + u = x^2$

$u' - u = -x^2$ A

(28) $u' - u = -x^2$

$y' + p(x) = q(x)$

$p(x) = -1$

$u = e^{\int -1\, dx}$

$u = e^{-x}$

$ue^{-x} = \int -x^2 e^{-x}$ $\Rightarrow ue^{-x} = -\int x^2 e^{+x}$

Using by parts we have,

$ue^{-x} = -1\left(e^x x^2 - 2xe^{-x} - 2e^{-x} + C\right)$

$u = x^2 + 2x + 2 + Ce^x$

but $u = y^{-1}$

B.

$$y = x^2 + 2x + 2 + ce^{-x}$$

$$y = (x^2 + 2x + 2 + ce^{-x})^{-1}$$

B

(4) $\begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 4 \\ e^{2t} \end{bmatrix}$

$x(t) = 2x + 2y + 4$

$y(t) = 2y + e^{2t}$

let $x = x_1$ $y = x_2$

$x_1' = 2x_1 + 2x_2 + 4$ — (1)

$x_2' = 2x_2 + e^{2t}$ — (2)

Diff (1) gives us

$x_1'' = 2x_1' + 2x_2'$ — (3)

Substitute (2) into (3)

$x_1'' = 2x_1' + 4x_2 + 2e^{2t}$

$x_1'' - 4x_2 = 2x_1' + 2e^{2t}$ — (4)

Rearrange (1)

$x_1' - 2x_2 = 2x_1 + 4$

$2x_1' - 4x_2 = 4x_1 + 8$ — (5)

(4) − (5)

$x'' - 2x_1' = 2x_1' + 2e^{2t} - 4x_1 - 8$

$x_1'' - 4x_1' + 4x_1 = 2e^{2t} - 8$ — (*)

Solve the y_c of the d.e

$m = 2, 2$ repeated roots

$y_c = Ge^{2t} + C_2te^{2t}$

$y_p = At^2e^{2t} + B$

$y_p' = 2Ate^{2t} + 2At^2e^{2t}$

$y_p'' = 4Ate^{2t} + 2Ae^{2t} + 4Ate^{2t} + 4At^2e^{2t}$

Substituting into (*)

$2Ae^{2t} + B = 2e^{2t} - 8$

$A = 1$ $B = -8$

$y_p = t^2e^{2t} - 8$

$x_1(t) = Ge^{2t} + C_2te^{2t} + t^2e^{2t} - 8$

Different $x_1(t)$ and the subst into (1) to obtain $x_2(t)$

$x_2(t) = te^{2t}$

$\therefore$ Soln $= \begin{bmatrix} x_1p(t) \\ x_2p(t) \end{bmatrix} = \begin{bmatrix} -2 + te^{2t} \\ te^{2t} \end{bmatrix}$

= 1

(31) $(2x+2y) + (2x+2y)\frac{dy}{dx} = 0$

A

For exactness $M_y = N_x$

$M_y = 2 \quad N_x = 2$

$\int (2x+2y)\,dx = x^2 + 2xy$

$\int (2x+2y)\,dy = 2xy + y^2$

$\Rightarrow x^2 + 2xy + y^2 = 0$

$\phi = x^2 + 2xy + y^2$

for $\phi\, y(0) = 0$

$c = 0$

$\phi = x^2 + 2xy + y^2$

$y(1) = ?$

$0 = 1 + 2y + y^2$

$= (y+1)(y+1)$

$y = -1$

B

(38) C

This is bcos there is a product of the dependent variable y & its der

(40) Multiplying thru by ye^x gives

$(e^x \sin y - 2y\sin x)\,dx$

$(e^x \cos y + \cos x)\,dy$

$\int M\,dx = \int (e^x \sin y - 2y\sin x)\,dx$

$\Rightarrow e^x \sin y + 2y\cos x$

$\int N\,dy = \int (e^x \cos y + \cos x)\,dy$

$= e^x \sin y + \phi$

$\Rightarrow e^x \sin y + 2y\cos x = C$

D

35. A

36. B

C

B

B

(41) $y'' - 4y' - 12y = te^{4t}$

$y_c = Ge^{6x} + Ge^{-2x}$

$y_p = (At + B)e^{4t}$

$y_p = Ate^{4t} + Be^{4t}$

$y_p' = Ae^{4t} + 4Ate^{4t} + 4Be^{4t}$

$y_p'' = 4Ae^{4t} + 4Ae^{4t} + 16Ate^{4t} + 16B$

$\left[\frac{d^2y}{dx^2} + \frac{dy}{dx}\right]^{1/5 \times 5} = \left[\frac{dy}{dx} + \sin^2 x\right]^5$

$\left(\frac{d^2y}{dx^2} + \frac{dy}{dx}\right) = \left(\frac{dy}{dx} + \sin^2 x\right)^5$

degree refers to the highest power of highest derivative

$4Ae^{4t} - 12Ate^{4t} - 12Be^{4t} = te^{4t}$

$A = \frac{-1}{12}$

$4Ae^{4t} - 12Be^{4t} = 0$

$B = -1/36$

$y_p = \left(\frac{-1}{12}t - \frac{1}{36}\right)e^{4t}$

$y_p = \frac{-1}{36}(3t+1)e^{4t}$ C

(4) $\frac{dy}{dx} = \frac{x^2}{x^2+1}$

$\int dy = \int \frac{x^2}{x^2+1}\,dx$

$\int dy = \int \left(1 - \frac{1}{x^2+1}\right)dx$

$\int dy = \int 1\,dx - \int \frac{1}{x^2+1}\,dx$

$y = x - \tan^{-1}x + C$

$y(0) = 0$

$C = 0$

$y = x - \tan^{-1}x$.

$f(1) =$

$y = 1 - \frac{1}{4}\pi$

$y = \frac{1}{4}(4-\pi)$ D

(45) $y' + \frac{y}{x} = \sin x$

$y' + p(x)y = q(x)$

$p(x) = \frac{1}{x}$

$\mu = e^{\int p(x)\,dx}$

$\mu = e^{\ln x}$

$\mu = x$

$xy = \int x \sin x$

using by parts.

$y = -\cos x + \frac{1}{x}\sin x + C$

$f(\pi) = 1$

$1 = -1 + C$

$C = 2$

$y = -\cos x + \frac{1}{x}\sin x$

$f(\frac{1}{2}\pi)$

$y = -\cos\left(\frac{1}{2}\pi\right) + \frac{2}{\pi}\sin\frac{\pi}{2}$

$y = 0 + \frac{2}{\pi}$

$y = \frac{2}{\pi}$ B

(46) $(D^2-2D+1)y = 2xe^x$

$(D+1)(D-1)y = 2xe^x$

$(D-1)y = \frac{1}{D-1} \cdot 2xe^x$

$a=1$

$(D-1)y = e^x \int e^{-x} \cdot 2xe^x$

$= e^x \int 2x$

$= e^x(x^2+C)$ — Rx

$y = \frac{1}{D-1} \cdot Rx$ $\quad a=1$

$y = e^x \int e^{-x}(x^2e^x + Ce^x)$

$y = e^x \int (x^2 + C)\,dx$

$y = e^x\left(\frac{x^3}{3} + Cx + C\right)$

$y = (Cx+C)e^x + \frac{x^3e^x}{3}$

D

$\Rightarrow y_p = t^2e^{-4t}(Bt^2+Ct+D)$ A

(49) $(D-4)x + 3y = 0$

$-6x + (D+7)y = 0$

Representing in matrix form we have

$$\begin{pmatrix} D-4 & 3 \\ -6 & D+7 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using Cramer's rule

$$\begin{vmatrix} D-4 & 3 \\ -6 & D+7 \end{vmatrix} x = \begin{vmatrix} 0 & 3 \\ 0 & D+7 \end{vmatrix}$$

$(D-4)(D+7) + 18 = 0$

$D^2+7D-4D-28+18=0$

$D^2+3D-10=0$

$D = 2, -5$

$x(t) = Ce^{2t} + Ce^{-5t}$

$x'(t) = 2Ce^{2t} - 5Ce^{-5t}$ — (1

$Dx - 4x + 3y = 0$

Substitute (1) & $Dx(t)$ to find y

$y(t) = \frac{2}{3}Ce^{2t} + 3Ce^{-5t}$

$y(t) = Ce^{2t} + Ce^{-5t}$

D

(47) A $y_c = Ce^{-4t} + Cte^{-4t}$

$y_p = At^2e^{-4t} + (Bt^2+Ct+D)t^2e^{-4t}$

Factorize t^2e^{-4t}

$= t^2e^{-4t}(A + Bt^2 + Ct + D)$

NB. A constant plus another constant gives a constant

$C + A = P$

(47) $y'' + 2y' + 5y = 0$

$-1 \pm 2i$

$y_c = e^{-x}(A\cos 2x + B\sin 2x)$

E

51. B

53. $xy\frac{dy}{dx} = x+1$

$\int y\,dy = \int \frac{(x+1)}{x}\,dx$

Hence seperable.

54. B homogenous with a degree of 2.

55. 0 This under the orthogonality of sine & cos

56. 2π C

58. $\pm 1, 0$

The singularity of a d.e refers to the highest coefficient that turns the highest derivative to zero.

59. $|x(x-3)|$ where $x=3$

$|3(3-9)| = 24$

$\therefore > 4$

$|x-3| > 4$

60. $y = \sum_{n=0}^{\infty} a_n x^{n+m}$

Refer to frobenius expansion more info.

(48) $\begin{bmatrix} e^{2t} & e^{-2t} & 0 \\ e^{2t} & 0 & e^{-t} \\ e^{2t} & e^{-2t} & e^{-t} \end{bmatrix}$

Determinant

$e^{2t}(-e^{-3t}) - e^{-2t}(e^{t} - e^{t}) + 0$

$-e^{-t} \qquad 0 \quad + 0$

$-e^{-t}$

(42) $\frac{\partial(f,x)}{\partial x} = 3x^2y^2e^{y}$

$f(x) = x^3y^2e^{y^3/3} + h(y)$

$\frac{\partial f(x)}{\partial y} = x^3(2y\,e^{y^3/3} + y^2e^{y^3/3})$

$= 2$

$\Rightarrow 2x^3 y e^{y^3/3} + x^3 y^4 e^{y^3/3} + h(y')$

$\int h(y') = 0$

$f(x) = x^3 y^2 e^{y^3/3} + 0$

$f(x) = c$

$c = x^3 y^2 e^{y^3/3}$

B

19

② $y' + \frac{1}{\ln t(t-3)} y = \frac{t}{\ln t}$

Points of discontinuity are (1,3)

A

③ D

④ $y' - 2y = e^{2t}$

$y_c = Ce^{2t}$ $\quad y_p = Ate^{2t}$

$y_p' = Ae^{2t} + 2Ate^{2t}$

Sub

$Ae^{2t} + 2Ate^{2t} - 2Ate^{2t} = e^{2t}$

$A = 1$

$y_p = te^{2t}$

G.S $y = Ce^{2t} + te^{2t}$

$y(0) = 2$

$2 = C$

$y = 2e^{2t} + te^{2t}$

$y = (2+t)e^{2t}$

⑤ B

⑥ $ty' - 2y = 2\cos 2t$

$y' - \frac{2}{t}y = \frac{2\cos t}{t}$

$p(x) = -\frac{2}{t}$

$\mu = e^{-\int \frac{2}{t} dt}$

$\mu = e^{-2\ln t}$ $\quad = \mu = e^{\ln}$

$\mu = \frac{1}{t^2}$

A

⑦ $(\sin t)y' + \frac{1}{t-1} y = \frac{t}{t+1}$

$y' + \frac{1}{\sin t(t-1)} y = \frac{t}{t+1}$

Points of discontinuity

$(0, 1)$

⑧ For fundamental soln the Wro should not be zero.

$y_1(t) = e^t$ $\quad y_2(t) = e^{t+1}$

$\begin{vmatrix} e^t & e^{t+1} \\ e^t & e^{t+1} \end{vmatrix}$

$e^t(e^{t+1}) - e^t(e^{t+1}) = 0$

Hence it can't be a fundamental

$\begin{vmatrix} \sin t & \sin 2t \\ \cos t & 2\cos 2t \end{vmatrix}$

~~$2\sin t\cos t - \cos t$~~

$2\sin t\cos 2t - \cos t \sin 2t \neq 0$

Hence are fundamental soln

B

using Abel's

$\int p(x)$

$1 = Ce^{-6\int \frac{1}{t-5}}$

$= Ce^{-6\ln t}$

$1 = Ce^{\ln(t-5)^{-6}}$

$\Rightarrow Ce^{\ln(t-5)^{-6}}$

$\mu = (t-5)^{-6}$

$y = Ce^{2x}$

$y(0) = 7$

$7 = C$

$= 2Ce^{2x}$

$y(0) = 7$

$\lambda = 2C$

$\therefore$ the Cim

$\lambda = 14$

up Simul

using abel's theorem the w.

$$y'' + \frac{t}{(t-5)}y' + \frac{\sec^2(t)}{t-5}y = 0$$

$$u = ce^{-\int p(x)\,dx}$$

$$= ce^{-\int \frac{1}{t-5}\,dt}$$

$$u = ce^{-\ln(t-5)^{-6}}$$

$$\Rightarrow ce^{\ln(t-5)^{-6}}$$

$$\ln u = -\ln(t-5)^{-6} c \ln e$$

$$u = \frac{c}{(t-5)^6}$$

$$\lambda = -6 \pm 4i$$

$$e^{-t}(A\cos 4t + B\sin 4t)$$

B

$$y = Ge^{2x} + Ge^{-3x}$$

$$y(0) = 7$$

$$7 = G_1 + G_2$$

$$1 = 2G_1 - 3G_2$$

$$y(0) = 7$$

$$\lambda = 2G - 3G$$

the $\lim_{t\to\infty} y(t) = 0$

$\lambda = 14$

Simultaneously we get

$y = 7e^{2x}$

Differentiating it and substituting it into the original equation gives zero. Which proves that it's a solution to the d.e.

(14) $y'' - 4y = e^{2t}\cos t$

$y_c = G_1 e^{2t} + G_2 e^{-2t}$

$y_p = e^{2t}t\,(A\cos t + B\sin t)$

$y_p = Ate^{2t}\cos t + Bte^{2t}\sin t$

C

(15) $\int_0^\infty e^{-st} f(t)\,dt$

$f(t) = te^{4t}$

$\int_0^\infty te^{4t}\cdot e^{-st}$

$\int_0^\infty t(e^{4t-st}) = \int_0^\infty te^{t(4-s)}$

C

(16) $F(s) = \frac{2s-1}{s^2+4s+5}$

$\frac{2s}{s^2+4s+5} - \frac{1}{s^2+4s+5}$

$s^2+4s+5 = (s+2)^2+1$

$\frac{2s}{(s+2)^2+1} \quad -\frac{1}{(s+2)^2+1}$

$\downarrow$

$\frac{2(s+2-2)}{(s+2)^2+1} \Rightarrow \frac{2(s+2)}{(s+2)^2+1} \quad \frac{-2}{(s+2)^2+1}$

$\downarrow \qquad \downarrow$

$2e^{-2t}\cos t \quad 2e^{-2t}\sin t$

Rewriting them we have

$2(e^{-2t}\cos t - 2e^{-2t}\sin t) - e^{-2t}\sin t$

$2e^{-2t}\cos t - 4e^{-2t}\sin t - e^{2t}\sin t$

$f(\pi/2) =$

$2e^{-2(\pi/2)}\cos \pi/2 - 4e^{-2(\pi/2)}\sin \pi/2 + e^{2(\pi/2)}\sin \pi/2$

$0 - 4e^{-\pi}(1) - e^{\pi}(1)$

$-5e^{-\pi}$ A

(17) $\frac{2s^2+20s+36}{s^3+6s^2+9s}$

$\Rightarrow \frac{2s^2+20s+36}{s(s+3)^2}$

Using partial fractions.

$\frac{A}{s} + \frac{B}{s+3} + \frac{C}{(s+3)^2}$

$\Rightarrow 2s^2+20s+36 = A(s+3)^2(s+3) + B(s(s+3)) + C(s(s+3))$

Solving the partial fraction gives us

$\frac{4}{s} + \frac{2}{s+3} + \frac{2}{(s+3)^2}$

$\downarrow \quad \downarrow \quad \downarrow$

$4 \quad -2e^{-3t} + 2te^{-3t}$ D

(18) $y' - 5y = t^3e^{2t}$

$L(y) = sL(y) - y(0)$

$5L(y) = 5L(y)$

$L(t^3e^{2t}) = \frac{3!}{s^{3+1}}$ Introducing shift

$= \frac{3!}{(s-2)^4} = \frac{6}{(s-2)^4}$

$sL(y) - y(0) - 5L(y) = \frac{6}{(s-2)^4}$

$sL(y) - 5L(y) = \frac{6}{(s-2)^4} + t$

$L(y)(s-5) = \frac{6}{(s-2)^4} + t$

$L(y) = \frac{6}{(s-5)(s-2)^4} + \frac{4}{(s-5)}$

(18) $2\sin 3t + 3\cos 4t$

$\downarrow$

$\frac{2(3)}{s^2+3^2} + \frac{3s}{s^2+4^2}$

$\Rightarrow \frac{6}{s^2+9} + \frac{3s}{s^2+16}$ A

(19) $c=1$ (Calculator work)

OR

$4m^2 + 4m + c = 0$

(22) $F(s) = \frac{s-1}{s^2+3s+2} = \frac{s-1}{(s+1)(s+2)}$

Using partial fractions

$s-1 = A(s+2) + B(s+1)$

$A=-2 \quad B=3$

$= \frac{-2}{s+1} + \frac{3}{s+2}$

$= -2e^{-t} + 3e^{-2t}$ A

(23) $f'' = s^2L(y) - s(y)(0) - y'(0$

$L(y) = \frac{s-1}{s^2-s-2}$

$= s^2\left(\frac{(s-1)}{s^2-s-2}\right) - s(-1) - 3$

$= \frac{s^2(s-1)}{s^2-s-2} + s - 3$

B

$\frac{2}{3}\int_0^\infty \frac{t\,dt}{2e^{(s-4)t}}$

$\frac{2}{3}\times\frac{1}{2}\int_0^\infty \frac{t}{e^{(s-4)t}} = \int_0^\infty te^{-(s-4)t}$

$te^{-st}\cdot e^{4t} \Rightarrow \int_0^\infty te^{4t}\cdot e^{-st}$

te^{4t}

(24) $9y'' + 6y' + y = 0$

$9m^2 + 6m + 1 = 0$

$m = -1/3, -1/3$

$y_c = Ge^{-1/3t} + Gte^{-1/3t}$

$y(0) = 0$

$y_p = Ae^{3t} + (Bt+C)\cos 5t + \dfrac{}{\sin 5t}$

$0 = C_1 e^{1/3} + C_2 e^{-1/3}$

$C_1 e^{-1/3} = -C_2 e^{-1/3}$

$C_1 = \dfrac{-C_2 e^{-1/3}}{e^{-1/3}}$

$C_1 = -C_2$

$y_c' = -\frac{1}{3} C_1 e^{-\frac{1}{3}t} + C_2 e^{-\frac{1}{3}t} + \frac{1}{3} C_2 t e^{-\frac{1}{3}t}$

$y(0) = 1$

$1 = -\frac{1}{3} C_1 e^{-1/3} + C_2 e^{-1/3} - \frac{1}{3} C_2 e^{-1/3}$

$1 = \frac{1}{3} C_2 e^{-1/3} + C_2 e^{-1/3} - \frac{1}{3} C_2 e^{-1/3}$

$1 = C_2 e^{-1/3}$

$C_2 = e^{1/3}$ $\quad C_1 = -C_2$

$C_1 = -e^{1/3}$

$y_c = -e^{1/3} e^{-\frac{1}{3}t} + e^{1/3} t e^{-\frac{1}{3}t}$

A

(25) $e^{3t} - 4t\cos 5t = y'' + y$

$y_c = A\cos t + B\sin t$

$y_p = Ae^{3t} + (Bt+C)[D\cos 5t + E\sin 5t]$

(26) $\int_0^4 e^{-3t} f(t) + \int_4^\infty e^{-t}$

$\Rightarrow \int_0^4 e^{-st}(0) = 0$

$\int_4^\infty e^{-st} \cdot e^{2t} = \int_4^\infty e^{-(s-2)t}$

$\Rightarrow \int_4^\infty e^{-(s-2)t} \, dt$

(27) B

(28) $y' - y = e^{-5t}$

$y' = sL(y) - y(0) - L(y) =$

$= y(0) = 2$

$\Rightarrow s[L(y)] - 2 - L(y) = \frac{1}{s+5}$

$(s-1)L(y) = \frac{1}{s+5} + 2$

$L(y) = \frac{1}{(s+5)(s-1)} + \frac{2}{(s-1)}$

(29) y'

$y_c =$

$y_p =$

$1 - \delta =$

$$\frac{1+2s+10}{(s+5)(s-1)} = \frac{2s+11}{(s+5)(s-1)}$$

B

29) $y'' + 5y' + 6y = t^2 + 1$

$y_c = Ge^{-2t} + Ge^{-3t}$

$y = At^2 + Bt + C$

$G.S = Ge^{-2t} + Ge^{-3t} + At^2 + Bt + C$

B

30) $y'' + 8y' + 16y = 0$

$y_c = Ge^{-4t} + G_2te^{-4t}$

$y(0) = 1$

$1 = G + 0$ — (1)

$y' = -4Ge^{-4t} - 4G_2te^{-4t} + G_2e^{-4t}$

$y'(0) = 0$

$0 = -4G - 4(0) + G_2$

$0 = -4G + G_2$ — (2)

Solving (1) & (2) simultaneously gives us

$G_2 = 4G$

$G_2 = 4$

$y_c = e^{-4t} + 4te^{-4t}$

It is a Riccati's eqn

Term $y_1(x) = \sin x$ always appears in y soln

Hence $y = \frac{2\sec x}{-2\tan x} + \sin x$

No need to add.

Linearly independent.

Refer to handout page 83

functions involving cos & sin will always be independent.

$\frac{dr}{dt} = -4rt$

$\int \frac{dr}{r} = \int -4t\,dt$

$\ln r = -2t^2 + C$ — (1)

when $r = r_0$ $t = 0$

$\ln r_0 = C$

Substitute into (1)

$\ln r = -2t^2 + \ln r_0$

$\ln r - \ln r_0 = -2t^2$

$\ln \frac{r}{r_0} = -2t^2$

$\frac{r}{r_0} = e^{-2t^2}$

$r = r_0 e^{-2t^2}$

(6) $y' + y = \sin x$

$y_c = Ce^{-x}$

$y_p = x^1[A\sin x + B\cos x]$

$y_p = A\sin x + B\cos x$

$y_p' = A\cos x - B\sin x$

Substituting into original eqn

$A\sin x - B\sin x + A\cos x + B\cos x = \sin x$

$A - B = 1$

$A\cos x + B\cos x = 0$

$A + B = 0$

$A = \frac{1}{2}$ $B = -\frac{1}{2}$

$y = Ce^{-x} + \frac{1}{2}\sin x - \frac{1}{2}\cos x$

(7) $xy\frac{dy}{dx} = x^2 + y^2$ — (1)

$xy\,dy = (x^2 + y^2)\,dx$

$y = vx$ — (1) $dy = v\,dx + x\,dv$ — (2)

Substitute both into eqn (1)

$x^2 v\,dy = (x^2 + x^2v^2)\,dx$

$v\,dy = (1 + v^2)\,dx$

$v(v\,dx + x\,dv) = (1 + v^2)\,dx$

$v^2 dx + xv\,dv = dx + v^2 dx$

$xv\,dv = dx$

$v\,dv = \frac{1}{x}dx$

$\int v\,dv = \int \frac{1}{x}dx$

$\frac{v^2}{2} = \ln x + c$

$\frac{v^2}{2} = 2\ln|x| + c$ $\quad v = \frac{y}{x}$

$\frac{y^2}{2x^2} = \ln|x| + c$

$y^2 = 2x^2(\ln|x| + c)$

(8) $\frac{dy}{dx} = \frac{y}{x}$

$x\,dy = y\,dx$

$0 = y\,dx - x\,dy$

(9) $9y\frac{dy}{dx} + 4x = 0$

$\int 9y\,dy + \int 4x\,dx = 0$

$\frac{9y^2}{2} + \frac{4x^2}{2} = c$

$y^2 + \frac{4}{9}x^2 = \frac{2}{9}c$

(10) $y' = -y^2 + xy + 1$

$y_1(x) = x$ (NB: Riccati)

$y = \frac{1}{z} + x$

$\frac{dy}{dx} = -\frac{1}{z^2}\frac{dz}{dx} + 1$

$y^2 = \left(\frac{1}{z} + x\right)^2 = \frac{1}{z^2} + \frac{2x}{z} + x^2$

Substituting into original eqn

$-\frac{1}{z^2}\frac{dz}{dx} + 1 = -\frac{1}{z^2} + \frac{2x}{z} - x^2 + \frac{xz+1}{z}$

$-\frac{1}{z^2}\frac{dz}{dx} = -\frac{1}{z^2} - \frac{x}{z}$

$-\frac{dz}{dx} = -1 - xz$

$z' - xz = 1$

$y' + p(x)y = q(x)$

$p(x) = -x$

$\mu = e^{\int p(x)\,dx} = e^{\int x}$

$\mu = e^{-x^2/2}$

$e^{-x^2/2} = \int e^{-x^2/2} = \frac{1}{-x}e^{x^2/2} + c$

$e^{-x^2/2} = \frac{e^{x^2/2}}{-x} + c$

$z = \frac{1}{-x} + ce^{+x^2/2}$

$z = \frac{1}{y-x}$

$y - x = \frac{1}{-1/x + ce^{x^2/2}}$

$-\frac{1}{x} + Ce^{x^2/2}$

$\frac{-1 + Ce^{x^2/2}}{x}$ ⇒

$y - x = \frac{x}{Ce^{x^2/2} - 1}$

$y = \frac{x}{Ce^{x^2/2} - 1} + x$

$(D-3)(D+3)y = e^{3x}$

$(D+3)y = \frac{1}{D-3} \cdot e^{3x}$

$D-3 = D-a \quad a = 3$

$= e^{3x} \int e^{-3x} \cdot e^{3x} \, dx$

$e^3 \int 1 \, dx$

$y(D+3) = e^{3x}(x + C)$

$D+3 = D-a \quad a = -3$

$y = e^{-3x} \int e^{3x}(xe^{3x} + C)$

$y = e^{-3x} \int xe^{6x} + Ce^{3x}$

$y = e^{-3x}\left[\frac{xe^{6x}}{6} - \frac{e^{6x}}{36} + \frac{Ce^{6x}}{6} + C_2\right]$

$y = \frac{xe^{3x}}{6} - \frac{e^{3x}}{36} + \frac{Ce^{3x}}{6} + C_2 e^{-3x}$

(12) $y' + \frac{2}{x}y = -x^3y^2$

$y^{-2}y' + \frac{2}{x}y^{-1} = -x^3$

Let $y^{-1} = v \quad \frac{dv}{dx} = -1y^{-2}\frac{dy}{dx}$

$\frac{-1}{1}\frac{dv}{dx} + \frac{2}{x}v = -x^3$

$\frac{dv}{dx} - \frac{8}{x}v = x^3$

$p(x) = -\frac{8}{x} \quad u = e^{\int p(x)\,dx}$

$u = e^{-\int \frac{8}{x}} = e^{-8\ln x}$

$u = \frac{1}{x^8}$

$\frac{v}{x^8} = \int x$

$\frac{v}{x^8} = \frac{x^2}{2} + C$

$v = \frac{x^{10}}{2} + Cx^8$

$y^{-1} = \frac{x^{10}}{2} + Cx^8$

(15) $y'' + y = \tan x$

$y_c = A\cos x + B\sin x$

$y_1 = \cos x \quad y_2 = \sin x$

using variation of parameters.

$y_p = u_1 y_1 + u_2 y_2$

$u_1 = -\int \frac{y_2 f(x)}{W} \qquad W(y_1, y_2) = 1$

$u_1 = -\int \sin x \tan x \Rightarrow \int \frac{\sin x \sin x}{\cos x}$

(16) $u_1 = -\int \frac{\sin^2 x}{\cos x}$

from trig identities $\sin^2 x + \cos^2 x = 1$

$u_1 = -\int \frac{1-\cos^2 x}{\cos x} = \int \frac{1}{\cos x} - \int \cos x$

$u_1 = -\left[\int \sec x - \int \cos x\right]$

$= -\left[\ln(\sec x + \tan x) - \sin x\right]$

$u_1 = -\ln(\sec x + \tan x + \sin x$

$u_2 = \int \frac{y_1 f(x)}{W} = \int \frac{\cos x \cdot \sin x}{\cos}$

$u_2 = -\cos x$

$y_p = \cos x\left[-\ln(\sec x + \tan x) + \sin x\right]$

$- \sin x \cos x$

(16) $x y' + xy - 4y = 0$

$y_1(x) = x^3$

$y_2 = y_1 \int \frac{e^{-\int p(x) dx}}{y_1^2}$

$p(x) = \frac{1}{x}$

$\mu = e^{-\int \frac{1}{x} dx} =$

$\mu = \frac{1}{x}$

$y_2 = x^3 \int \frac{\frac{1}{x}}{x^6} = \int x$

$y_2 = x^3\left[\frac{1}{-6} x^{-6}\right]$

$y_2 = -\frac{x^{-3}}{6}$

$G.S = C_1 x^3 + C_2 \frac{x^{-3}}{6}$

(17) $m y\, dx = n x\, dy$

$\int \frac{m}{x} dx = \int \frac{n}{y} dy$

$m \ln x = n \ln y + \ln c$

$m \ln x - \ln c = n \ln y$

$\frac{x^m}{c} = y^n$

$W(y_1, y_2)$

$y_1 = 2x^2 \qquad y_2 = x^4$

$$\begin{vmatrix} 2x^2 & x^4 \\ 4x & 4x^3 \end{vmatrix}$$

$4x^3(2x^2) - 4x^5 = \underline{\underline{4x^5}}$

$e^{y'+y} = x$

$y' + y = \ln x$

$y' = \ln|x| - y$

$\frac{dy}{dx} = (\ln|x| - y)$

$dy = (\ln|x| - y)\,dx$

$dy - (\ln|x| - y)\,dx = 0$

$9y''' + 12y'' + 4y' = 0$

$e^{mx}(9m^3 + 12m^2 + 4m) = 0$

$m(9m^2 + 12m + 4) = 0$

$m = 0 \qquad -2/3 \qquad -2/3$

$y = Ge^0 + G_2 e^{-2/3} + G_3 x e^{-2/3}$

$y = G + G_2 e^{-2/3} + G_3 x e^{-2/3}$

(22) $y'' + 5y' + 6y = e^{-2x}$

$y_c = Ge^{-3x} + G_2 e^{-2x}$

$y_p = Ax^k e^{-2x}$

$y_p = Axe^{-2x}$

$y_p' = Ae^{-2x} + 2Axe^{-2x}$

$y_p'' = 2Ae^{-2x} + 2Ae^{-2x} + 4Axe^{-2x}$

$= 4Ae^{-2x} + 4Axe^{-2x}$

$-Ae^{-2x} - 6Axe^{-2x} + 6Axe^{-2x} = e^{-2x}$

$A = \underline{\underline{-1}}$

$y_p = -xe^{-2x}$

(23) $y''' + y'' = 3e^x + 4x^2$

$m^3 + m^2 = 0$

$m(m^2 + m) = 0$

$m = 0, 0, -1$

$y_c = G + G_2 x + G_3 e^{-x}$

$y_p = Ae^x + Ax^2 + Bx + C$

$y_p' = Ae^x + 2Ax + B$

$y_p'' = Ae^x + 2A$

$y_p''' = Ae^x$

$2Ae^x + 2A = 3e^x + 4x^2$

$A = \underline{\underline{\frac{3}{2}}}$

$G.S = G + G_2 x + G_3 e^{-x} + \frac{3}{2}e^x$

(24) $(D+3)(D+3)y = xe^{-3x}$

$D+3 = D-a \quad a=-3$

$(D+3)y = e^{-3x}\int e^{3x}(xe^{-3x})$

$= e^{-3x}\int x^2$

$= e^{-3x}(\frac{x}{2}+c)$ — Rx

$y = \frac{1}{D+3} \cdot Rx^2 \qquad a=-3$

$y = e^{-3x}\int e^{3x}(\frac{x^2}{2}e^{-3x} + ce^{-3x})$

$= e^{-3}\int (x+c)\,dx$

$= e^{-3x}\left[\frac{x^3}{6} + cx + c_2\right]$

$y = \frac{x^3e^{-3}}{6} + c_1xe^{-3x} + c_2e^{-3x}$

(25) $y'' + 4y' + 4y = e^{-2x}$

$y_c = c_1e^{-2x} + c_2xe^{-2x}$

$y(0) = 1$

$1 = c_1$

$y'(0) = -1$

$y_c' = -2c_1e^{-2x} - 2c_2xe^{-2x} + c_2e^{-2x}$

$-1 = -2c_1 + 0 + c_2$

$-1 = c_2$

$y_c = e^{-2x} + xe^{-2x}$

$y = e^{-2x}(1+x)$

(18) $(6x+4y+1)dx + (4x+2y+2)dy = 0$

$6x+4y=-1 \qquad 4x+2y=-2$

$x = X - \frac{3}{2} \qquad y = Y+2$

Substitute in eqn

$(6x+4Y)dx + (4x+2Y)dy = 0$

let $y = xv$

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

$dy = vdx + xdv$

$(6x+4xv)dx + (4x+2xv)(vdx+xdv) = 0$

$(6+4v)dx + (4+2v)vdx + x(4+2v)dv = 0$

$6dx + 4vdx + 4vdx + 2v^2dx + 4xdv + 2vxdv = 0$

$(2v^2+8v+6)dx + (4x+2vx)dv = 0$

$\frac{xdv}{dx} = \frac{-2v^2-8v-6}{4+2v}$

$\int \frac{4+2v}{-(2v^2+8v+6)}dv = \int \frac{1}{x}dx$

$\frac{1}{2}\ln(2v^2+8v+6) = -\ln x + \ln c$

$(2v^2+8v+6)^{1/2} = -xc$

$\ln(2v^2+8v+6)^{1/2} = \ln\frac{c}{x}$

$(2v^2+8v+6) = \frac{c}{x^2}$ —

Substitute $Y = y-2$ & $X = x+\frac{3}{2}$

$v = \frac{Y}{X}$ into eqn (*)

$\left(\frac{y-2}{x+3/2}\right)^2 + 8\left(\frac{y-2}{x+3/2}\right) + 6 = \frac{c}{(x+3/2)^2}$

Dividing thru by $(x+3/2)^2$

$(y-2)^2 + 8(y-2)(x+3/2) + 6(x+3/2)^2 = c$

At $y(1/2) = 3$

$c = 42$

Expanding it we have

$4y^2 - 16y + 16 + 16(xy + \frac{3}{2}y - 2x - 3) + 12x^2 + 36x + 27 = 84$

$4y^2 - 16y + 16xy + 24y - 32x - 48 + 12x^2 + 36x = 41$

$4y^2 + 8y + 16xy + 4x + 12x^2 = 89$

$xyy' = 2y^2 + 4x^2$

Let $y = xv$

$\frac{dy}{dx} = v + x\frac{dv}{dx}$

$dy = v\,dx + x\,dv$

$v\frac{dy}{dx} = (2x^2v^2 + 4x^2)$

$v\,dy = (2x^2v^2 + 4x^2)dx$

$v\,dy = (2v^2 + 4)\,dx$

$(v\,dx + x\,dv) = (2v^2 + 4)\,dx$

$v^2dx + xv\,dv = (2v^2dx + 4dx$

$xv\,dv = v^2dx + 4dx$

$xv\,dv = (v^2+4)dx$

$\int \frac{v}{v^2+4}\,dv = \int \frac{1}{x}\,dx$

$\frac{1}{2}\ln v^2+4 = \ln x + \ln c$

$\ln(v^2+4)^{1/2} = \ln xc$

$(v^2+4)^{1/2} = xc$

$v^2 + 4 = x^2c$

$v^2 = x^2c - 4$

$\frac{y^2}{x^2} = (x^2c - 4)$

$\frac{y^2}{x^2} = (x^2c - 4)$

$\frac{y}{x} = (x^2c - 4)^{1/2}$

$y = x\sqrt{x^2c - 4}$

NB: IVP shd ve been given to find c.

But if c is assumed to be 2 the our ans is D.

(9) None of the above.

(10) Repeated question. Same as 2013 que 31

(5) $\sin 2\pi\left(\frac{1}{2}\right) = 0$

(6) $\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} x$

Eigen values are given by

$\lambda^2 - \text{trace}\lambda + \det = 0$

$\lambda^2 - 6\lambda + 5 = 0$

$\lambda_1 = 5 \quad \lambda_2 = 1$

Eigen vector $\lambda_1 = 5$ $[A - I\lambda] = 0$

$$\begin{bmatrix} 2-5 & 1 \\ 3 & 4-5 \end{bmatrix} \begin{matrix} v_1 \\ v_2 \end{matrix} = \begin{bmatrix} -3 & 1 \\ 3 & -1 \end{bmatrix} \begin{matrix} v_1 \\ v_2 \end{matrix} = 0$$

$$\begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \qquad 3v_1 - v_2 = 0$$

$$v_2 = 3v_1$$

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$\lambda_1 = 1$ $[A - I\lambda] = 0$

$$\begin{bmatrix} 2-1 & 1 \\ 3 & 4-1 \end{bmatrix} \begin{matrix} v_1 \\ v_2 \end{matrix} = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} \begin{matrix} v_1 \\ v_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{matrix} 0 \\ 0 \end{matrix} \qquad v_1 + v_2 = 0$$

$$v_2 = -v_1$$

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$= G_1 v_1 e^{\lambda_1 t} + G_2 v_2 e^{\lambda_2 t}$

$$G\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + G_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{5t}.$$

(12) $\lambda^4 + 3\lambda^2 - \lambda + 2 = 0$

(15) $\begin{bmatrix} 2 & 1 \\ a & 2 \end{bmatrix}$ $\lambda_1 = 1$ $\lambda = 3$

$\lambda^2 - \text{trace}\lambda + \det = 0$

$\lambda^2 - 4 + (4 - a) = 0$

$4 - a = 3 \times 1$

$4 - 3 = a$

$a = 1$

OR

(16) $\begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix} x \qquad x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda^2 - \text{trace}\lambda + \det A = 0$

$\lambda^2 - 3\lambda - 10 = 0$

$\lambda_1 = 5 \quad \lambda_2 = -2$

Eigen vector for $\lambda_1 = 5$

$$\begin{bmatrix} 4-5 & 2 \\ 3 & -1-(5) \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \qquad (A - \lambda I)v = 0$$

$$\begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \Rightarrow \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \qquad v_1 - 2v_2 = 0$$

$$v_1 = 2v_2$$

$\begin{pmatrix} 2v_2 \\ x_2 \end{pmatrix} = v_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$x_1 = -2$

$(A - \lambda I)v = 0$

$\begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix} \begin{matrix} v_1 \\ v_2 \end{matrix} = 0 \Rightarrow \begin{pmatrix} 3 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{matrix} 0 \\ 0 \end{matrix}$

$\begin{pmatrix} 3 & 1 \\ 0 & 0 \end{pmatrix} \begin{matrix} v_1 \\ v_2 \end{matrix} = \begin{pmatrix} 1 & 1/3 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$

$3v_1 + \frac{1}{3}v_2 = 0$

$v_1 = -\frac{1}{3}v_2$

$\begin{pmatrix} -1/3 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

G.S $= G v_1 e^{\lambda_1 t} + G_2 v_2 e^{\lambda_2 t}$

$= G_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{5t} + G_2 \begin{pmatrix} -1 \\ 3 \end{pmatrix} e^{-2t}$

Using IVP $= x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$x: \quad 1 = 2G_1 e^{5t} + G_2 e^{-2t}$

$1 = 2G_1 - G_2$ ——— (1)

$y: \quad -1 = G_1 e^{5t} + 3G_2 e^{-2t}$

$-1 = G_1 + 3G_2$ ——— (2)

Solving (1) & (2) simultaneously

we have

$G_1 = \frac{2}{7} \qquad G_2 = \frac{-3}{7}$

$x(t) = \frac{4}{7}e^{5t} + \frac{3}{7}e^{-2t}$

$y(t) = \frac{2}{7}e^{5t} + \frac{9}{7}e^{-2t}$

$\frac{1}{7}\begin{bmatrix} 4e^{5t} & 3e^{-2t} \\ 2e^{5t} & -9e^{-2t} \end{bmatrix}$

(17) $x' = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} x + \begin{bmatrix} 4 \\ e^{2t} \end{bmatrix}$

$x_1' = 2x_1 + 2x_2 + 4$ ——— (1)

$x_2' = 2x_2 + e^{2t}$ ——— (2)

NB: $x' = 2x + 2y + 4$, $y' = 2y + e^{2t}$, let $x = x_1$, $y = x_2$

Differentiating (1) gives us

$x_1'' = 2x_1' + 2x_2' +$ ——— (3)

Substitute (2) into (3)

$x_1'' = 2x_1' + 4x_2 + 2e^{2t}$

$x_1'' - 4x_2 = 2x_1' + 2e^{2t}$ ——— (4)

Eliminate x_2 using (1)

$x_1' - 2x_2 = 2x_1 + 4$

$2x_1' - 4x_2 = 4x_1 + 8$ ——— (5)

eqxn(2) - eqxn(1)

$x'' - 4x' = 2x_1' + 2e^{2t}$

$-(2x_1' - 4x_2 = 4x_1 + 8)$

$x'' - 2x_1' = 2x_1' + 2e^{2t} - 4x_1 - 8$

$x'' - 4x_1' + 4x_1 = 2e^{2t} - 8$

Solving for y_c we have

$y_c = Ge^{2t} + Cte^{2t}$

$y_p = At^2e^{2t} - B$

$y_p' = 2Ate^{2t} + 2At^2e^{2t}$

$y_p'' = 2Ae^{2t} + 4Ate^{2t} + 4Ate^{2t} + 4At^2e^{2t}$

Substituting y_p'' & y_p' & y_p Give us

$2Ae^{2t} - 4B = 2e^{2t} - 8$

$A = 1 \quad B = 2$

$y_p = t^2e^{2t} - 2$

G.S: $x(t) = Ge^{2t} + Gte^{2t} + t^2e^{2t} - 2$

$x'(t) = 2Ge^{2t} + Ge^{2t} + 2Gte^{2t} + 2te^{2t} + 2t^2e^{2t}$

Substituting $x'(t)$ and $x(t)$ into eqxn (1) gives

$x_2(t) = te^{2t}$

G.S $= \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

$= \begin{bmatrix} t^2e^{2t} - 2 \\ te^{2t} \end{bmatrix}$

$= \begin{bmatrix} -2 + t^2e^{2t} \\ te^{2t} \end{bmatrix}$

(18) $f(t) = \begin{cases} 0 & t < 1 \\ t^2 & t \geq 1 \end{cases}$

$\int_0^1 e^{-st} f(t)\,dt$ where $f(t) =$

$\int_0^1 e^{-st}(0)\,dt$ NB: Whenever integrate and a constant, it can ignored.

$\int_0^\infty e^{-st} f(t)\,dt \quad f(t) = t^2$

$\int_0^\infty e^{-st} t^2$ Using by part we we

$\left[\frac{t^2e^{-st}}{-s} - \frac{2te^{-st}}{s^2} - \frac{2e^{-st}}{s^3}\right]$

$\infty \to 0$

$-\left[\frac{t^2e^{-st}}{-s} - \frac{2te^{-st}}{s^2} - \frac{2e^{-st}}{s^3}\right]$

at $f(0)$ we have terms of t

$= \frac{e^{-s}}{s} + \frac{2e^{-s}}{s^2} + \frac{2e^{-s}}{s^3}$

$f(1) = \frac{e^{-1}}{1} + \frac{2e^{-1}}{1^2} + \frac{2e^{-1}}{1^3}$

$= 5e^{-1}$

$\frac{1}{s} - \frac{s+2}{s^2+4}$

$\frac{s}{s^2+4} + \frac{2}{s^2+4}$ and $\frac{1}{s} = 1$

$\cos 2t + \sin 2t$

$1 - (\cos 2t + \sin 2t)$

Inserting $g(1)$ in radians we have

$1 - (-0.4161 + 0.9093)$

$0.5068 \approx 0.50$

$xy^2 \frac{dy}{dx} = y^3 + x^4$

$y = xu \quad \frac{dy}{dx} = u + x\frac{du}{dx}$

$dy = u\,dx + x\,du$

substituting ① & ② into the D.E

$x^3 u^2 dy = (y^3 + x^7)dx$

$x^3 u^2 dy = (u^3x^3 + x^4)dx$

$u^2(u\,dx + x\,du) = u^3 dx + x\,dx$

$u^3 dx + u^2 x\,du = u^3 dx + x\,dx$

$u^2 x\,du = x\,dx$

$\int u^2 du = \int dx$

$\frac{u^3}{3} = x + c$

$u^3 = 3(x+c)$

$\frac{y^3}{x^3} = 3(x+c)$

$y^3 = x^3 3(x+c)$

$y = x\sqrt[3]{3(x+c)}$

(22) Since they are exact, we integrate $\int Dy$ & $\int Dx$

$\int (3x^2y - 3y)\,dx \quad \| \quad \int (x^3 - 3x + 2y)\,dy$

$x^3y - 3xy + y^2$

$x^3y - 3xy$

$x^3y - 3xy + y^2 = c$

(23) $x'' - 5x + 4y = 0$
$y'' + 4x - 5y = 0$

For y-term Let
$x_1 = y$
$x_1' = y' = x_2$
$x_2' = y''$

For x-term Let
$x_3 = x$
$x_3' = x' = x_4$
$x_4' = x$

$x'' = 5x + 4y$
$x'' = 5x_3 + 4x_1$
$x_1'' = x_4'$

$x_4' = 5x_3 + 4x_1$
$y'' = -4x + 5y$
$y'' = x_2'$
$x_2' = -4x_3 + 5x_1$

$x_1' = x_2$
$x_2' = -4x_3 + 5x_1$
$x_3' = x_4$
$x_4' = 5x_3 - 4x_1$

(24) $L[y''] = s^2L(y) - sy(0) - y$
$L(y^2) = s^2L(y) - 4s - 8$

$L[y'] = sL(y) - y(0)$
$= sL(y) - 4$

$7L(y) =$

$2s^2L(y) - 8s - 16 + sL(y) - 4 + 7($
$2s^2(L(y) + sL(y) + 7L(y = 8s +$
$L(y)(2s^2 + s + 7) = 8s + 2$
$L(y) = \frac{8s + 20}{2s^2 + s + 7}$

(25) $\begin{bmatrix} 1 & 5 \\ 2 & 4 \end{bmatrix} x + \begin{bmatrix} 1 \\ te^{-t} \end{bmatrix}$

$x_1' = x_1 + 5x_2 + 1$ — (1)
$x_2' = 2x_1 + 4x_2 + te^{-t}$ — (2)

Diff (2) gives
$x_1'' = x_1' + 5x_2'$ — (3)
$x_1'' = x_1' + 5(2x_1 + 4x_2 + te^{-t})$
$x_1'' = x_1' + 10x_1 + 20x_2 + 5te^{-t}$
$x_1'' - 20x_2 = x_1' + 10x_1 + 5te^{-t}$

Eliminate x_2 by multiply (1) by
$4x_1' - 20x_2 = 4x_1 + 4$ — (5)

Eqn (4) − (1)

$x_1'' - 4x_1' = x' + 10x_1 + 5te^{-t} - 4x_1 + 4$

$x_1'' - 5x_1' - 6x_1 = 5te^{-t} + 4$

$y_c = Ge^{4t} + Se^{-t}$

$y_p = (At + B)te^{-t} + C$

$e^{-} \left[\frac{5}{4}e^{-2}\right] = \frac{5}{4}$

(28) $x_1'(t) = x_1 + 3x_2 + 2e^t$ — (1)

$x_2'(t) = x_2 + 2te^t$ — (2)

Diff (1)

$x_1'' = x_1' + 3x_2' + 2e^t$ — (3)

$x'' - 3x_2' = x_1' + 2e^t$

$x'' = x_1' + 3x_2 + 6te^t + 2e^t$

Substitute (2) in (3)

$x'' - 3x_2 = x_1' + 6te^t + 2e^t$ — (4)

$x_1' - 3x_2 = x_1 + 2e^t$ — (1)

Eqn (4) − (1)

$x'' - x_1' = x_1' - x_1 + 6te^t$

$x'' - 2x_1' + x_1 = 6te^t$

$y_c = Ge^t + Gte^t$

$y_p = t^2(At + B)e^t$

$y_p = At^3e^t + Bt^2e^t$

$y_p' = 3At^2e^t + At^3e^t + 2Bte^t + Bt^2e^t$

$y_p'' = 6Ate^t + 3At^2e^t + 3At^2e^t + Ae^t t^3 + 2Be^t + 2Bte^t$

(1) $\frac{2s-3}{s^2+9} \Rightarrow \frac{2s}{s^2+3^2} - \frac{3}{s^2+3^2}$

$\downarrow$

$2\cos 3t - \sin 3t$

(2)

for $g(s)$

$2(4) - 0$

$\frac{2}{}$

$\left[\frac{e^{-s}}{s} + \frac{2e^{-s}}{s^2} + \frac{2e^{-s}}{s^3}\right]$

$e^{7}f(2)$

replace $s = 2$

$\left[\frac{e^{-2}}{2} + \frac{2e^{-2}}{4} + \frac{2e^{-2}}{8}\right]$

Subst y_p'', y_p' & y_p into eqn

We have

$6At e^t + 2Be^t = 6te^t$

$A = 1$

$B = 0$

$y_p = At^3e^t$

$y_p = t^3e^t$

$x_1(t) = Ge^t + Gte^t + t^3e^t$

$x_1'(t) = Ge^t + Ge^t + Gte^t + 3t^2e^t + t^3e^t$

From eqn ①

$3x_2(t) = x_1'(t) + 2e^t$

$x_2(t) = \frac{G}{3}e^t + t^2e^t - 2e^t$

NB: A constant divided by constant gives constant

$\frac{G}{3} = G$

$x(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$0 = G - 2 \qquad G = 0$

$G = 2$

$x_2(t) = 2e^t + t^2e^t - 2e^t$

$x_2(t) = t^2e^t$

$x_1(t) = 2te^t + t^3e^t$

$$\begin{bmatrix} 2t + t^3 \\ t^2 \end{bmatrix} e^t$$

$xy' = y + \sqrt{x^2 - xv^2}$

Let $y = xv$

$xy' = vx + x\sqrt{1-v^2}$

$y' = v + \sqrt{1-v^2}$

$v + x\frac{dv}{dx} = v + \sqrt{1-v^2}$

$\frac{dv}{\sqrt{1-v^2}} = \frac{dx}{x}$

$\sin^{-1} v = \ln x + C$

$\sin^{-1}\left(\frac{y}{x}\right) = \ln x + C$

$C = 0$

$\frac{y}{x} = \sin(\ln x)$

$y = x \sin \ln(x)$

(3) $y' = v + x\frac{dv}{dx}$

(2) $\frac{dy}{dx} + y = \frac{\cos x}{e^x}$

$\mu = e^{\int 1}$ $\mu = e^x$

$e^x = \int \cos x$

$y = \frac{\sin x}{e^x} + \frac{C}{e^x}$

$y(0) = 1$

$C = 1$

$y = \frac{\sin x}{e^x} + \frac{1}{e^x}$

$y(\pi)$

$y = 0 + \frac{1}{e^\pi}$

$y = \frac{1}{e^\pi}$

(31) $y'' + y' - 2y =$

$y(t) = e^{xt}$ $y_2(t) = e^t$

$-2, 1$

$(m+2)(m-1) = 0$

$m^2 - m + 2m - 2 = 0$

$m^2 + m - 2 = 0$

$y'' + y' - 2y = 0$

(32) $y' - 6y = te^{6t}$

$y_c = Ce^{6t}$

$y_p = (At + B)te^{6t}$

$y_p = At^2e^{6t} + Bte^{6t}$

$y_p' = 2Ate^{6t} + 6At^2e^{6t} + Be^{6t} + 6Bte^{6t}$

$y_p' - 6y_p = 2Ate^{6t} + Be^{6t} = te^{6t}$

$2Ate^{6t} = te^{6t}$

$A = \frac{1}{2}$

$y_p = \frac{1}{2}t^2e^{6t}$

G.S: $Ge^{6t} + \frac{1}{2}t^2e^{6t}$

(33) For fundamental solutions their wronskian should not be zero

cost	$-9\sin t$
sint	$+9\cos t$

$9\cos^2 t + 9\sin^2 t = 1$

$W \neq 0$ Hence fundamental

$2e^{-t}$	$4e^{-t+3}$
$2e^{-t}$	$-4e^{-t+3}$

$-8e^{-2t+3} + 8e^{-2t+3} = 0$

Hence not a fundamental soln

(34) B

(35) $y' - \frac{(t+2)}{t}y = \frac{2\tan(2t)}{t}$

$\mu = e^{\int p(t)\,dt}$ $\quad p(t) = -\left(\frac{t+2}{t}\right)$

$= 1 + \frac{2}{t}$

$\int 1 + \int \frac{2}{t} = (t + 2\ln t)$

$\mu = (t + 2\ln t)$ $\quad \mu = e^{t + \ln t^2}$

$\mu = e^{-t} \cdot e^{\ln t^2}$

$\ln \mu = \ln e^{-t} \cdot \ln e^{\ln t^2}$

$\ln \mu = -t + \ln t^2$ $\quad \mu = \frac{1}{e^t t^2}$

$\mu = e^{-t} \cdot t^2$

$\mu = e^{-t}t^2$ $\quad \mu = \frac{1}{e^t t^2}$

(36) D

(37) $C_1e^t + C_2e^{-2t}$

$b = C_1 + C_2$ —— (1)

$2 = C_1 - 2C_2$ —— (2)

$b - 2 = 3C_2$ $\quad b - C_1 = C_2$

$C_2 = \frac{b-2}{3}$ $\quad b - C_1 = \frac{b-2}{3}$

$C_1 = \frac{3b}{3} - \frac{b-2}{3} = 4b$

$b - \frac{b-2}{3} = C_1$ ——

$C_1 = \frac{2b-2}{3}$

$\frac{3b-b-2}{3} = C_1$

$y = c_1 e^{t} + c_2 e^{-2t}$

$y = \frac{2b-2}{3} e^{t} + \frac{b-2}{3} e^{-2t}$

$\lim_{t \to \infty}$

$\infty \quad 0$

$\infty + 0 \Rightarrow \infty$

∴ Therefore there is no value for b with limit 0

5) For linearly dependence $f_1(t) = f_2(t)$

Hence $\lambda = 0$

1) $ty'' + y' + ty = 0$

$y'' + \frac{1}{t} y' + y = 0$

Using Abel's theorem

$\mu = ce^{-\int p(t)\,dt}$

$\mu = ce^{-\int \frac{1}{t} dt}$

$\mu = ce^{-\ln t}$

$\mu = ce^{-\ln t}$

$\mu = \ln ce^{-\ln t}$

$\mu = c - \ln t$

$\mu = ct^{-1}$

(40) $y'' + \frac{2}{t} y' - \frac{5}{t^2} e^{2t} y = 0$

$\mu = ce^{-\int \frac{2}{t} dt}$

$\mu = ce^{-2\ln t}$

$\mu = ce^{-2\ln t}$

$\ln \mu = \ln ce^{-2\ln t}$

$\ln \mu = c(-2\ln t$

$\mu = ct^{-2}$

$\mu = \frac{c}{t^2}$ $\mu = W$

$W = \frac{c}{t^2}$

$W(y_1, y_2)(1) = 3$

$3 = \frac{c}{1}$ $c = 3$

$W = \frac{3}{t^2}$

(41) From the principle of superposition $c_1 v_1 + c_2 v_2$ is also a solution.

$y_1 = 2e^{2t} + te^{2t}$

$y_2 = e^{5t} + te^{2t}$

$c_1 v_1 + c_2 v_2$

$2e^{2t} + te^{2t} - e^{5t} + te^{2t}$

$2e^{2t} - e^{5t} + 2te^{4t}$

$\Rightarrow \frac{1}{s-1} + \frac{1}{(s-2)^2} + \frac{(s-3)}{(s-3)^2+1}$

(42) $4y'' + y = 0$

$\pm \frac{1}{2}i$

$y(t) = C_1 \cos\frac{1}{2}t + C_2 \sin\frac{1}{2}t$

(43) $te^t = (At+B)e^t$

$y_c = e^{2t} + te^{2t}$

$\cos 2t = C\cos 2t + D\sin 2t$

$t+2 = At + B$

$\Rightarrow (At+B)e^t + C\cos 2t + D\sin 2t + Et + F$

(44) D Refer to handout.

(45) e^t | te^{4t}

$\downarrow$ | $\downarrow$

$\frac{1}{s-1}$ | $\frac{1!}{s^2} \Rightarrow \frac{1}{(s-2)^2}$

$e^{3t}\cos(4t)$

$\frac{s}{s^2+4^2}$ $\rightarrow \frac{(s-3)}{(s-3)^2+4^2}$

(46) $\lambda_1 = 4+3i$ $\lambda_2 = 4-$

$v_1 \begin{pmatrix} 1 \\ -i \end{pmatrix}$ $v_2 \begin{pmatrix} 1 \\ i \end{pmatrix}$

$C_1 v_1 e^{\lambda}$

$C_1 \begin{pmatrix} 1 \\ -i \end{pmatrix} e^{4+3i} \Rightarrow C_1 \begin{pmatrix} 1 \\ -i \end{pmatrix} e^{4t} \cdot e$

From euler $e^{3it} = \cos 3t + i\sin$

$C_1 \begin{pmatrix} 1 \\ -i \end{pmatrix} e^{4t} \cos 3t + i\sin 3t$

$C_1 \begin{pmatrix} \cos 3t + i\sin 3t \\ -i\cos 3t + \sin 3t \end{pmatrix} e^{4t}$

$C_2 \begin{pmatrix} 1 \\ i \end{pmatrix} e^{4-3i} = \begin{pmatrix} 1 \\ i \end{pmatrix} e^{4t} \cdot e^{-3it}$

$C_2 \begin{pmatrix} 1 \\ i \end{pmatrix} e^{4t} \cos 3t - i\sin 3t$

$C_2 \begin{pmatrix} \cos 3t - i\sin 3t \\ i\cos 3t + \sin 3t \end{pmatrix} e^{4t}$

Picking the real and imagin part we have

$C_1 \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} e^{4} + C_2 \begin{pmatrix} \pm\sin 3t \\ -\cos 3 \end{pmatrix}$

D

(48) $y' - \frac{10}{t}y = t^2 e^{2t}$

$p(x) = -\frac{10}{t}$

$\mu = e^{\int p(t)\,dt}$

$= e^{-\int \frac{10}{t}} = e^{-10\ln t} = e^{\ln t^{-10}}$

$\ln \mu = \ln e^{\ln t^{-10}}$

$\ln \mu = \ln t^{-10}$

$\mu = \frac{1}{t^{10}}$

(49) $y'' + 2y' + y = 0$

$y_c = C_1 e^{-t} + C_2 t e^{-t}$

$\lim_{t \to \infty} C_1 e^{-\infty} + C_2 \infty e^{-\infty}$

$0 \qquad 0$

All approach 0

B

$y' = \frac{\sin 2t}{\cos y}$

$\cos y \frac{dy}{dt} = \sin 2t$

$\int \cos y\,dy = \int \sin 2t\,dt$

$\sin y = -\frac{1}{2}\cos 2t + C$

$y = \sin^{-1}\left(-\frac{1}{2}\cos 2t + C\right)$

A

(52) Differentiate option A P.C and substituting them into the D.E gives D except D. Hence it's not a soln to the D.E

(53) $y'' + 4y' + 3y = 4e^{-2t}$

$y_c = C_1 e^{-t} + C_2 e^{-3t}$

$y_p = Ae^{-2t}$

$y_p' = -2Ae^{-2t}$

$y_p'' = +4Ae^{-2t}$

$4Ae^{-2t} - 8Ae^{-2t} + 3Ae^{-2t} = 4e^{-2t}$

$-Ae^{-2t} = 4e^{-2t}$

$A = -4$

$y_p = -4e^{-2t}$

$G.S = C_1 e^{-t} + C_2 e^{-3t} - 4e^{-2t}$

B

(54) $y''' - 9y' = 0$

$m = 0, -3, 3$

G.S $= Ae^{0} + Be^{-3t} + Ce^{3t}$

$y = A + Be^{-3t} + Ce^{3t}$

$2e^{-t}\cos 2t - \frac{1}{2}e^{-t}\sin 2t$ $\quad$ $\frac{\pi}{4}\sin 2t$

$2e^{-t}\cos 2t - \frac{3}{2}\sin 2t$

$2e^{-t}\cos 2t - \frac{3}{2}e^{-t}\sin 2t$

D

(56) $y'' = 5y' - 3y$

Let $y' = x_1$

$\frac{dy}{dt} = \frac{dx_1}{dt} = x_2$

$\frac{d^2x_2}{dt^2} = 5x_2 - 3x_1$

$x_1' = x_2$

$x_2' = 5x_2 - 3x_1$

B

(57) $\frac{2s-1}{s^2+2s+5} = \frac{2s-1}{(s+1)^2+2^2}$

$\frac{2s}{(s+1)^2} \quad \frac{-1}{(s+1)^2+2}$

$\frac{2(s-1+1)}{(s+1)^2+2^2} \Rightarrow \frac{2(s+1)}{(s+1)^2+2^2} \quad \frac{-1}{(s+1)^2+2^2}$

$\downarrow \qquad \downarrow$

$2e^{-t}\cos 2t \quad - \quad \frac{1}{2}\sin 2t$

$\frac{-1}{(s+1)^2+2} \Big| = \sin\sqrt{2}t$

$-1 \pm i$

(58) $y'' = -y' + 2y$

Let $y = x_1$

$\frac{dy}{dx} = \frac{dx_1}{dt} = x_2 \qquad \frac{d^2x_2}{dt^2} = -x_2$

$x_1' = x_2$

$x_2' = -x_2 + 2x_1$

C

(59) $\begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$

$x_1'(t) = -x_1 - x_2$ — (1)

$x_2'(t) = x_1 - x_2$ — (2)

Diff (1) we have

$x_1' = -x_1' - x_2'$

$x_1' = -x_1' - x_1 + x_2$

$x_1'' - x_2 = -x_1' - x_1$

$x_1' + x_2 = -x_1$

$x'' + x_1' = -x_1' - 2x_1$

$x'' + 2x_1' + 2x_1 = 0$

$y_c = e^{-t}(\cos t + \sin t)$

$x(t) = e^{-t}(\cos t + \sin t)$

$x_1(t) = e^{t}(A\cos t + B\sin t)$

$x_1(t) = A\cos t\, e^{t} + Be^{t}\sin t$

$x_1'(t) = e^{t}A\sin t - Ae^{t}\cos t - Be^{t}\sin t + Be^{t}\cos t$

$x_2(t) = -x_1' - x_1$

$= Ae^{t}\sin t + Ae^{t}\cos t + Be^{t}\sin t - Be^{t}\cos t - Ae^{t}\cos t - Be^{t}\sin t$

$= Ae^{t}\sin t - Be^{t}\cos t$

$x_2(t) = e^{t}(A\sin t - B\cos t)$

For IVP $x(0) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

$2 = A$ $\quad \rightarrow \quad 2 = e^{0}(A\cos 0 + B\sin 0)$

$-1 = -B$ $\quad\quad 1 = e^{0}(0 - B)$

$B = +1$

$x_1(t) = (2\cos t + \sin t)e^{-t}$

$x_2(t) = (2\sin t + \cos t)e^{-t}$

$x_1(t) = e^{-t}(2\cos t + \sin t)$

$x_2(t) = e^{-t}(2\sin t - \cos t)$

$$\begin{bmatrix} 2\cos t + \sin t \\ -\cos t + 2\sin t \end{bmatrix} e^{-t}$$

(c) $y'' - y' - 2y = 4x^2$

$m^2 - m - 2 = 0$

$m = 2, -1$

$y_c = Ce^{2x} + Ce^{-x}$

$y_p = Ax^2 + Bx + C$ — (1)

$y_p' = 2Ax + B$ — (2)

$y_p'' = 2A$ — (3)

Substituting (1), (2), (3) in d.e

$2A - 2Ax - B - 2Ax^2 - 2Bx - 2C = 4x^2$

$-2Ax^2 = 4x^2$

$-2A = 4$

$A = -2$

$-2Ax - 2Bx = 0$

$2A + 2B = 0$

$2B = +4$

$B = 2$

$2A - B - 2C = 0$

$-4 - 2 - 2C = 0$

$-2C = 6$

$C = -3$

$y_p = -2x^2 + 2x - 3$

G.S $= Ce^{2x} + Ce^{-x} - 2x^2 + 2x - 3$

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1. C
2. C
3. A
4. $e^{\int g(x)\,dx}$ (Repeated Quest)

(5) $\sum_{n=1}^{\infty} \frac{t^n}{n!}$ is the series expansion for e^x

$f(s) = 6e^s = \frac{6}{s-1}$

(6) $f(t) = e^{at}\cos(at)$

$\cos(at) = \frac{s}{s^2+a^2}$

$e^{wt}\cos(at) = \frac{s-w}{(s-w)^2+a^2}$

(7) $t^n e^{at}$

$t^n = \frac{n!}{s^{n+1}}$

$e^{at}t^n = \frac{n!}{(s-a)^{n+1}}$

(8) $\int_1^5 (3+2t-4t^2)\,\delta(t-3)\,\delta t$

$3+2(-3)-4(-3)^2$

-27 //

(9) $L(\cosh 2t\ \delta(t))$

$\delta(t-0)$ $a=0$

We put 0 into cosh

$\cosh(0) = 1$ //

$f(s)\,e^{-as}$

$1 \cdot e^0 \Rightarrow 1$ //

(12) $\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/4} = K\frac{d^2y}{dx^2}$

$\left[1+\left(\frac{dy}{dx}\right)^2\right]^3 = K\left[\frac{d^2y}{dx^2}\right]^4$

Degree = 4

(13) $ye^x = x+c$

$y = xe^{-x} + ce^{-x}$

$y' = -xe^{-x} + e^{-x} - ce^{-x}$

$y' = e^{-x} - (xe^{-x} + ce^{-x})$

$y' = e^{-x} - y$

$y' + y = e^{-x}$ Repeated

(10) $G(s) = \dfrac{86s - 18}{(s+3)(s-4)(5s-1)}$

$86s - 78 = \dfrac{A}{s+3} + \dfrac{B}{s-4} + \dfrac{C}{5s-1}$

$86s - 78 = A(s-4)(5s-1) + B(s+3)(5s-1) + C(s+3)(s-4)$

$s = 4$

$266 = B(7)(19)$

$266 = 133B$

$B = \underline{2}$

$s = -3$

$-336 = A(-7)(-16)$

$-336 = +112A$

$A = \underline{-3}$

$s = 5$

$352 = A(1)(24) + B(8)(24) + C(8)(1)$

$352 = -72 + 384 + 8C$

$40 = 8C$

$C = \underline{5}$

$\Rightarrow \dfrac{-3}{s+3} + \dfrac{2}{s-4} + \dfrac{5}{5s-1}$

$\downarrow$

$-3e^{-3t} + 2e^{+4t} + e^{1/5t}$

$\rightarrow \dfrac{5}{5}\left[\dfrac{1}{s-1/5}\right]$

(11) $\frac{25}{s^3(s^2+4s+5)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{}{s^2+4s+5}$

$25 = As^2(s^2+4s+5) + Bs(s^2+4s+5) + C(s^2+4s+5)$
$(Ds+E)(s^3)$

WAS DONE AGAIN ON THE LAST PAGE

$s = 0$

$25 = 5C$

$C = \underline{5}$

$= As^4 + 4As^3 + 5As^2 + Bs^3 + 4Bs^2 + 5Bs + Cs^2 + 4Cs + 5C$
$Ds^4 + Es^3$

$As^4 + Ds^4 = 0$ ⇒

$A + D = 0$ — s^4

$4As^3 + Bs^3 + Es^3 = 0$

$4A + B + E = 0$ — s^3

$5As^2 + 4Bs^2 + Cs^2 = 0$

$5A + 4B + C = 0$ — s^2

$5Bs + 4Cs = 0$

$5B + 4C = 0$ — s

But $C = 5$

$5B = -4C$

$B = \underline{-4}$

$5A + 4B + C = 0$

$5A - 16 + 5 = 0$

$A = \frac{11}{5}$

$A + B + E = 0$

$E = \frac{11}{5} + 4$

$E = \frac{9}{5}$

$A + D = 0$

$D = \frac{-11}{5}$

$\frac{11}{5s} + \frac{4}{s^2} + \frac{5}{s^3} + \frac{1}{5}\frac{11s-9}{s^2+4s+5}$

$\frac{1}{5}\left(\frac{11}{s} - \frac{20}{s^2} + \frac{25}{s^3} - \frac{11s-9}{s^2+4s+5}\right)$

$\frac{11}{s} = \boxed{11}$

$\frac{20}{s^2} = \boxed{20t}$

$\frac{25}{s^3} = \boxed{\frac{25t^2}{2}}$

$\frac{11s-9}{s^2+4s+5} = \frac{11s}{s^2+4s+5} = 11\frac{(s+2)-2}{(s+2)^2+1}$

$= 11\frac{(s+2)}{(s+2)^2+1} \rightarrow \boxed{11e^{-2t}\cos t}$

$\frac{-2}{(s+2)^2+1} = \boxed{-2e^{-2t}\sin t}$

$= \frac{1}{5}\left(11 - 20t - \frac{25}{2}t^2 - 11e^{-2t}\cos t - 2e^{-2t}\right)$

(15) $y' + ax = k$

$(D+a)\ x = k$

$y = e^{-at}\int e^{at} \cdot k\,dt$

(16) E

(17) $Ce^{-x} + Cxe^{-x}$

$Ae^{-x} + Bxe^{-x}$

$Ae^{-t} + Bte^{-x}$

(18) $y_c = Ae^{2x} + Be^{x}$

$y_p = Ae^{-x} + Be^{-2x}$

$y_p' = -Ae^{-x} - 2Be^{-2x}$

$y_p'' = Ae^{x} + 4Be^{-2x}$

Inserting y_p'', y_p' & y_p into the D.E

$e^{x} + 4Be^{-2x} + 3Ae^{-x} + 6Be^{-2x} + 2Ae^{-x} + 2Be^{-2x} + 6Ae^{-x} + 12Be^{-2x} = e^{-x} + e^{-2x}$

$A = \frac{1}{6} \quad B = \frac{1}{12}$

$y_p = \frac{1}{6}e^{-x} + \frac{1}{12}e^{-2x}$

$Ae^{x} + Be^{x}$

$Ae^{2x} + Be^{x} + \frac{1}{6}e^{-x}$

(21) $\begin{vmatrix} e^{x} & e \\ e^{x} & 2e^{2x} \end{vmatrix}$

$2e^{3x} - e^{3x} = \underline{\underline{e^{3x}}}$

(22) $y_c = C_1e^{2x} + C_2e^{x}$

$y_c' = 2C_1e^{2x} + C_2e^{x}$

$4 = 2C_1 + C_2$ —— (1)

$3 = C_1 + C_2$ —— (2)

Solving both simultaneously

$C_1 = 1 \quad C_2 = 2$

$y_p = Ae^{-t}$

$y_p' = -Ae^{-t} \quad y_p'' = Ae^{-t}$

$Ae^{-t} + 3Ae^{-t} + 2Ae^{-t} = e^{-t}$

$6Ae^{-t} = e^{-t}$

$A = \frac{1}{6}$

$y_p = \underline{\underline{\frac{1}{6}e^{-t}}}$

$G.S = e^{2t} + 2e^{t} + \frac{R}{6}^{-t}$

(23) $\frac{\lambda^2}{\lambda^4} = \lambda^{-2} = \underline{\underline{-2}}$

(24) Solve $2x+3y=3$ and $3x+2y=7$ Simultaneously

$x=3 \quad y=-1$

$x = X+3 \quad y = Y-1$

(25) Repeated question 20B que 22

(26) E

(27) $y'' + \frac{1}{2x} - \frac{3y}{2x^2} = 0$

$p(x) = \frac{1}{2x}$

$\mu = e^{\int p(x)\,dx}$

$\mu = e^{\frac{1}{2}\int \frac{1}{x}\,dx}$

$\mu = e^{\frac{1}{2}\ln x}$

$\mu = e^{\ln x^{1/2}}$

$\ln \mu = \ln x^{1/2} \ln e$

$\ln \mu = \ln x^{1/2} \quad \mu = \underline{\underline{\sqrt{x}}}$

(28) $y_c = A\cos x + B\sin x$

$y_p = u_1 y_1 + u_2 y_2$

$u_1 = -\int \frac{y_2 f(x)}{W}$

$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$

$\cos^2 x + \sin^2 x = 1$

$u_1 = -\int \frac{\sin x \cdot \sec x}{1}$

$= -\int \tan x = -(-\ln\cos x)$

$= \ln\cos x$

$u_2 = -\int \frac{\cos x \cdot \sec x}{1} = \int 1$

$= x$

$u_2 y_1 = \cos x (\ln\cos x$

$u_2 y_1 = \sin x (x)$

$y_p = \underline{\underline{\cos x (\ln\cos x) + x\sin x}}$

(29) $m\frac{d^2x}{dt^2} = ma$

$\frac{d^2x}{dt^2} = a \qquad a = -qx$

$\frac{d^2x}{dt} - qx = 0$

$m^2 - qm = 0$

$\pm 3i$

$A\cos 3t + B\sin 3t$

$4y'' + 9y = 0$

$4m^2 + 9 = 0$

$\pm 3/2 i$

$c_1 \cos 3/2 t + c_2 \sin 3/2 t$

(2) $\frac{dy}{dx} = x\sqrt{x^2+9}$ $\quad y(-4) = 0$

Let $x^2 + 9 = u$

$\frac{du}{dx} = 2x \Rightarrow \frac{du}{2x} = dx$

$dy = (x\sqrt{u})\,dx$

$dy = (x\sqrt{u})\frac{du}{2x}$

$\int dy = \frac{1}{2}\int \sqrt{u}\,du$

Seperable so we integrate both sides.

$y = \frac{1}{2}\left[u^{3/2} \cdot \frac{2}{3} + c\right]$

$y = \frac{1}{2}\left[\frac{2}{3}u^{3/2} + c\right]$

$y = \frac{1}{3}u^{3/2} + \frac{c}{2}$

∴ NB: A constant divided by a constant gives a constant. Let $\frac{c}{2} = \underline{\underline{c}}$

$y = \frac{1}{3}u^{3/2} + c$

$0 = \frac{1}{3}\left((-4)^2+9\right)^{3/2} + c$

$0 = \frac{1}{3}(125) + c$

$c = -\frac{1}{3}(125)$

$\therefore y = \frac{1}{3}(x^2+9)^{3/2} - \frac{1}{3}(125)$

$y = \frac{1}{3}\left[(x^2+9)^{3/2} - 125\right]$

(3) $\frac{3s+1}{(s-1)(s^2+1)}$

Using the method of partial fractions we have.

$\frac{3s+1}{(s-1)(s^2+1)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+1}$

$3s+1 = A(s^2+1) + Bs+C(s-1)$

$3s+1 = A(s^2+1) + Bs+C(s-1)$

$s=1 \qquad s=0$

$4 = 2A \qquad 1 = A + C$

$A = \underline{\underline{2}} \qquad C = \underline{\underline{1}}$

$3s+1 = As^2 + A + Bs^2 - B + C$

$3s+1 = 2s^2 + 2 + Bs^2 - B + s$

$2s - 2s^2 = Bs^2 - B$

$Bs^2 = -2s^2$

$B = -2$

$$\frac{2}{s-1} + \frac{1-2s}{s^2+1}$$

$$\Rightarrow \frac{2}{s-1} + \frac{1}{s^2+1} - \frac{2s}{s^2+1}$$

$\downarrow$ $\quad$ $\downarrow$

$2e^t \quad \sin t - 2\cos t$

$$2e^t + \sin t - 2\cos t$$

(4) $y' = 1 + x + y + xy$

$y' = (1+x) + y(1+x)$

$\frac{dy}{dx} = (1+x) + y(1+x)$

$\frac{1}{(1+x)}\frac{dy}{dx} = (1+y)$

$\frac{1}{(1+y)}\,dy = (1+x)\,dx$

$\int \frac{1}{(1+y)}\,dy = \int (1+x)\,dx$

$\ln|1+y| = x + \frac{x^2}{2} + C$

(5) $3xy' + y = 12x$

$y' + \frac{1}{3x}y = \frac{12x}{3x}$

$\mu = e^{\int p(x)\,dx}$

$p(x) = \frac{1}{3}x$

$\mu = e^{\int 1/3x}$

$\mu = e^{1/3\int 1/x} = e^{1/3 \ln x} = e^{1}$

$\ln \mu = \ln x^{1/3} \ln e$

$\mu = x^{1/3}$

$y(x^{1/3}) = 4x^{1/3}$

$yx^{1/3} = \int 4x^{1/3}$

$yx^{1/3} = 4\left[\frac{3}{4}x^{4/3} + C\right]$

$yx^{1/3} = 3x^{4/3} + C$

$y = 3x + Cx^{1/3}$

(6) $(3x^2y^3 + y^4)\,dx + (3x^3y^2 + y^3)\,dy = 0$

Test for exactness

$My = 9x^2y^2 + 4y^3$

$Nx = 9x^2y^2 + 4y^3$

hence we integrate our given

$\int (3x^2y^3 + y^4)\,dx = x^3y^3 + xy^4$

$\int (3x^3y^2 + y^4 + 4xy^3)\,dy$

$\Rightarrow x^3y^3 + \frac{y^5}{5} + xy^4 = F$

$x^3y^3 + \frac{y^5}{5} + xy^4 = c$

$5x^3y^3 + y^5 + 5xy^4 = 5c$

$5c = c$

$5x^3y^3 + y^5 + 5xy^4 = C$

(8) $y_1(t) = e^t \sin t$

$e^t \sin t$ → $e^t \cos t$

$e^t \sin t + e^t \cos t$ → $e^t \cos t - e^t \sin t$

$e^{2t}\sin^2 t$

$e^{2t}\sin t\cos t - e^{2t}\sin^2 t - e^{2t}\sin t\cos t - e^{2t}\cos^2 t$

$-e^{2t}\sin^2 t - e^{2t}\cos^2 t$

$-e^{2t}(\sin^2 t + \cos^2 t)$

$-e^{2t}(1) \Rightarrow \underline{\underline{-e^{2t}}}$

9. A

(10) $\frac{dy}{dx} + \frac{1}{2}y = 2+t$

$\frac{dy}{dx} + \frac{1}{2}y = (2+t)$

$P(x) = \frac{1}{2}$ $\mu = e^{\int P(x)\,dx}$

$\mu = e^{\frac{1}{2}\int 1\,dx} = \underline{\underline{e^{\frac{1}{2}x}}}$

$ye^{\frac{1}{2}x} = \int e^{\frac{1}{2}x}(2+t)$

$= \int 2e^{\frac{1}{2}x} + te^{\frac{1}{2}x}$

Solving $te^{\frac{1}{2}x}$ using by parts

$ye^{\frac{1}{2}} = [4e^{\frac{1}{2}t} + 2te^{\frac{1}{2}t} - 4e^{\frac{1}{2}t}$

(7) $\sin^2 3t =$

From trig identities

$\sin^2 t = \frac{1}{2}(1 - \cos 2t)$

$\sin^2 3t = \frac{1}{2}(1 - \cos 6t)$

$\frac{1}{2}\left(\frac{1}{s} - \frac{s}{s^2+6^2}\right)$

$\Rightarrow \frac{1}{2s} - \frac{s}{2(s^2+6^2)}$

$\frac{s^2+6^2 - s^2}{2(s^2+6^2)s} = \frac{36}{2(s^2+6^2)s}$

$\Rightarrow \frac{18}{s(s^2+36)}$ //

$ye^{1/2t} = 2te^{1/2t} + c$

$y = 2t + ce^{-1/2t}$

At the point (0, 2)

$t = 0 \quad y = 2$

$2 = 0 + c$

$c = 2$

$y = 2t + 2e^{-1/2t}$

(12) B

(13) B

(14) $\frac{dy}{dx} = \frac{y}{x^2}$

$\int \frac{1}{y} dy = \int \frac{1}{x^2} dx$

$\ln c + \ln y = -\frac{1}{x}$

$\ln yc = -1/x$

$yc = e^{-1/x}$

$y = ce^{-1/x}$ $y = ce^{1/x}$

Since $y = ce^{-1/x} + d$ is not a func

~~$c = 2$~~ ~~$d = 0$~~ at x.

None of ~~the~~ above

(15) $y'' + 3y' + 2y = \cos^2 x$

$y_c = m^2 + 3m + 2 = 0$

$m = -1, -2$

$y_c = c_1 e^{-x} + c_2 e^{-2x}$

$y_p = \cos^2 x$

From trig identities

$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$y_p = \frac{1}{2}[A + B\cos 2x + C\sin 2x]$

(16) $x' = x\tan t + \sin t$

$x' - x\tan t = \sin t$

$\mu = e^{\int p(x) dx}$

$p(x) = \tan t$

$\int \tan t = -\ln \cos x$

$\mu = e^{-\int \tan t}$

$\mu = e^{\ln \cos t}$

$\mu = \ln \cos t$

$\mu = \cos t$

$y_p' = 2B\sin 2x + 2C\cos 2x$

$y_p'' = -4B\cos 2x - 4C\sin 2x$

$-4B\cos 2x - 4C\sin 2x - 6B\sin 2x + 6C\cos 2x + 2A + 2B\cos 2x + 2C\sin 2x$

$2A - 2B\cos 2x + 6C\cos 2x - 2C\sin 2x - 6B\sin 2x = 1 + \cos 2x$

$2A = 1$

$A = \frac{1}{2}$

$-2B\cos 2x + 6C\cos 2x = \cos 2x$

$-2B + 6C = 1$ ——— (1)

$-2C\sin 2x - 6B\sin 2x = 0$

$-6B - 2C = 0$ ——— (2)

Solving (1) & (2) simultaneously gives us-

$B = \frac{-1}{20} \qquad C = \frac{3}{20}$

$y_p = \frac{1}{2}\left[\frac{1}{2} - \frac{1}{20}\cos 2x + \frac{3}{20}\sin 2x\right]$

$= \frac{1}{4} - \frac{1}{40}\cos 2x + \frac{3}{40}\sin 2x$

$y_p = \left[\frac{10 - \cos 2x + 3\sin 2x}{40}\right]$

(16) $x' = x\sin t + 2te^{-\cos t}$

$x' - x\sin t = 2te^{-\cos t}$

$p(x) = -\sin t$

$\mu = e^{-\int \sin t\, dt}$

$\mu = e^{-(-\cos t)}$

$\mu = e^{\cos t}$

$xe^{\cos t} = 2te^{-\cos t} \cdot e^{\cos t}$

$xe^{\cos t} = \int 2t$

$xe^{\cos t} = t^2 + c$

$x = (t^2 + c)e^{-\cos t}$

(11) $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$

$y_c = m^2 - 3m - 3 = 0$

$m = 3, -1$

$y_c = Ge^{3x} + Ge^{-x}$

$y_p = Ax + B + C + e^{2x}(Dx + E)$

*NB: Since B & C are constant $B + C = Q$ where Q is another constant.

$y_p = Ax + Q + Dxe^{2x} + Ee^{2x}$ ——— (1)

$y_p' = A + De^{2x} + 2Dxe^{2x} + 2Ee^{2x}$ ——— (2)

$y_p'' = 2De^{2x} + 2De^{2x} + 4Dxe^{2x} + 4Ee^{2x}$ ——— (3)

Substituting (1), (2) & (3) into the general eqn:

$2De^{2x} + 2De^{2x} + 4Dxe^{2x} + 4Ee^{2x} - 2A - 4Dxe^{2x} - 4Ee^{2x} - 3De^{2x} - 3Ax - 3Q - 3Dxe^{2x} - 3Ee^{2x} = 4x - 5 + 6xe^{2x}$

$De^{2x} - 2A - 3Ax - 3Q - 3Dxe^{2x} - 3Ee^{2x} = 4x - 5 + 6xe^{2x}$

$-3Ax = 4x$

$A = -\frac{4}{3}$

$-3Dxe^{2x} = 6xe^{2x}$

$-3D = 6$

$D = -2$

$-2A - 3Q = -5$

$2A + 3Q = 5$

$2\left(-\frac{4}{3}\right) + 3Q = 5$

$3Q = 5 + \frac{8}{3}$

$3Q = \frac{23}{3}$

$Q = \frac{23}{9}$ (4)

$2De \mid -3te^{-} = 0$

$2D \mid -3E = 0$

$-4 \mid 3E = 0$

$E = -\frac{4}{3}$

$$y_p = -\frac{4}{3}x + \frac{23}{9} + e^{2x}\left(-2x - \frac{4}{3}\right)$$

$$y_p = -\frac{4}{3}x + \frac{23}{9} - \left(2x + \frac{4}{3}\right)e^{2x}$$

(12) $\frac{dv}{dt} = 1000 - \frac{1}{40}v^2$

$\frac{dv}{1000 - \frac{1}{40}v^2} = dt$ } Seperable Hence we integrate both sides

$$\int \frac{1}{1000 - \frac{1}{40}v^2}\,dv = \int dt$$

$1000 - \frac{1}{40}v^2$

$\frac{40000 - v^2}{40} \Rightarrow \int \frac{40}{40\,000 - v^2} \Rightarrow 40\int \frac{1}{40{,}000 - v^2} = 40\int \frac{1}{200^2 - v^2}$

From hyperbolic integrals

$\int \frac{1}{a^2 - u^2} = \frac{1}{a}\tanh^{-1}\frac{u}{a} + C \Rightarrow 40\left[\frac{1}{200}\tanh^{-1}\frac{1}{200}v\right] = t + C$

$\frac{1}{5}\tanh^{-1}\frac{1}{200}v = t + c$

$\tanh^{-1}\frac{1}{200}v = 5t + C \quad v = 200\tanh 5t + C$

$v(0) = 5$

$= 200(1) + C$

$C = 200$

$v(0) = 5$

$v = 200 \tanh 5t + C$

$5 = 0 + C$

$C = 5$

$v = 200 \tanh 5t + C$

After a very long time : we take $t = 100$

$200 \tanh 5(100)$

$\downarrow$

$v = 200(1)$

$v = 200$ m/s There4 after a very long time it shll be moving at 200m/s . You can try for other va at t.

(4) $\cos(x+y) + 2x + (\cos(x+y) + 4y)y' = 0$

$[\cos(x+y) + 2x]dx + [\cos(x+y) + 4y]dy = 0$

Test for exactness.

$M_y = (-\sin(x+y)$ $N_x = -\sin(x+y)$.

$\int \cos(x+y) + 2x \, dx$ $\int \cos(x+y) + 4y \, dy$

$\downarrow$ $\downarrow$

$\sin(x+y) + x^2$ $\sin(x+y) + 2y^2$

$\Rightarrow \sin(x+y) + x^2 + y^2 = C$

MID-SEM 2014

① $(x+1)y'' - (x+2)y' + y = 0$ $\quad y_1(x) = x^3$

$y'' - \frac{(x+2)}{(x+1)} y' + y = 0$

$p(x) = -\frac{(x+2)}{(x+1)}$ $\quad y_2 = y_1 \int \frac{e^{-\int p(x)\,dx}}{y_1^2}$

$\mu = e^{-\int p(x)dx} = \mu = e^{\int \frac{x+2}{x+1}} = \int 1 + \frac{1}{x+1} \Rightarrow x + \ln x + 0$

$\mu = e^{x + \ln x + 1} \Rightarrow \mu = e^x \cdot e^{\ln x + 1}$

$\ln \mu = \ln e^x \cdot \ln e^{\ln x + 1}$

$\ln \mu = \ln(x+1) \cdot \ln e^x$

$\mu = (x+1)e^x$

$y_2 = x^3 \int \frac{(x+1)e^x}{x^6}$

$y_2 = x^3 \int (x^{-5} + x^{-6}) e^x$

$y_2 = \frac{x^3 \, \text{Euler}(x) - e^x(x^4 + x^3 + 2x^2 + 6x + 4)}{20x^2}$

② $\begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} X$

$X^2 - \text{trace}A + \det A = 0$

$\lambda = -1, -1$

Eigen vector $\lambda = -1 \quad (A - I\lambda)v = 0$

$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{matrix} 0 \\ 0 \end{matrix}$

$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{matrix} 0 \\ 0 \end{matrix}$

$v_1 + v_2 = 0 \quad v_1 = -v_2$

$v_2 = v_2 \quad \begin{pmatrix} +1 \\ -1 \end{pmatrix}$

Eigen vector $\lambda_2 = -1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Since it a repeated roots

$X = X_1 + X_2$

$X_1 = G_1 v_1 e^{\lambda t} = G\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$

$X_2 = C_2\left(v_1 t e^{\lambda t} + \omega e^{\lambda t}\right)$

for ω's eigen vectors we have

$(A - I\lambda)\omega = V_1$

$\begin{bmatrix} 0-(-1) & 1 \\ -1 & -2-(-1) \end{bmatrix}\begin{matrix} w_1 \\ w_2 \end{matrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}\begin{matrix} w_1 \\ w_2 \end{matrix} = \begin{matrix} 1 \\ -1 \end{matrix}$

$w_1 + w_2 = 1$

$-w_1 - w_2 = -1$

let $w_1 = 0$ then $w_2 = 1$

Eigenvector $= \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$X_2 = G_2\left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{-t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t}\right]$

G.

$G.S = G\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} + G_2\left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{-t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t}\right]$

③ $x_1(t) = Ge^{-t} + G_2 t e^{-t}$

$x_2(t) = -Ge^{-t} + G_2 t e^{-t} + G_2 e^{-t}$

$X(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$2 = G + \quad \text{——} (1)$

$1 = -G + G_2$

$G_2 = 3$

$G.S = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} + 3\left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{-t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t}\right]$

④ $(xy^2 + ax^2y)\,dx + (x^3 + x^2y)\,dy = 0$

For exactness

$My = Nx$

$My = 2xy + ax^2 \qquad Nx = 3x^2 + 2xy$

$2xy + ax^2 = 3x^2 + 2xy$

$a = 3$

(11) $A + D = 0$

$4A + B + E = 0$

$5A + 4B + C = 0$

$-5B + 4C = 0$

$B = \frac{-4 \times 5}{5}$

$B = -4$

$A = \frac{-C - 4B}{5} = \frac{-5 + 16}{5} = \frac{11}{5}$

$A + D = 0$

$D = \frac{-11}{5}$

$E = -\frac{24}{5}$

$\Rightarrow \frac{11}{5s} - \frac{4}{s^2} + \frac{5}{s^3} + \frac{(-11s - 24)}{5(s^2 + 4s + 5)}$

$\frac{1}{5}\left(\frac{11}{s} - \frac{20}{s^2} + \frac{25}{s^3} + \left(\frac{-11s - 24}{(s+2)^2 + 1}\right)\right)$

$\left(11 - 20t + \frac{25}{2}t^2 - 11e^{2t}\cos t + 22e^{2t}\sin t - 24e^{2t}\sin t\right)$

$11 - 20t + \frac{25}{2}t^2 - 11e^{-2t}\cos t - 2e^{-2t}\sin t$

$A = \frac{11}{5}$ $B = -4$ $C = 5$

$D = \frac{-11}{5}$ $E = \frac{-24}{5}$

$\frac{11}{s} = 11$ $\frac{20}{s^2} = 20t$

$\frac{25}{s^3} = \frac{25}{2}t^2$

$\frac{-11s}{(s+2)^2 + 1} = \frac{-11[(s+2) - 2]}{(s+2)^2 + 1}$

$-11\left[\frac{(s+2)}{(s+2)^2 + 1} \quad \frac{-2}{(s+2)^2 + 1}\right]$

$-11\left[e^{-2t}\cos t - 2e^{-2t}\sin t\right]$

$-11e^{-2t}\cos t + 22e^{-2t}\sin t$

$\frac{-24}{(s+2)^2 + 1} = -24e^{-2t}\sin t$

(20) $t(t-2)(t-4)y' + y = e^t$

$y' + \frac{y}{t(t-2)(t-4)} = \frac{e^t}{t(t-2)(t-4)}$

$t \in (2, 4)$

SECTION B

(a) i. $m_1 = 3 \quad m_2 = -5 \quad m_3 = 1$

for roots of multiplicity

$m_1 = 3 \quad m_2 = -5 \quad m_3 = -5 \quad m_4 = 1 \quad m_5 = 1 \quad m_6 = 1$

$y = c_1 e^{3x} + c_2 e^{-5x} + c_3 x e^{-5x} + c_4 x^2 e^{-5x} + c_6 e^{x} + c_7$

$y = c_1 e^{3x} + c_2 e^{-5x} + c_3 x e^{-5x} + c_4 e^{x} + c_5 x e^{x} + c_6 x^2 e^{x}$

(ii) for Cauchy - euler eqn

$y = c_1 x^3 + c_2 x^{-5} + c_3 x^{-5}(\ln x) + c_4 x^{4} + c_5 x(\ln x) + c_6 x (\ln x)^2$

(b) i. variation of parameters

ii. undetermined co-efficients

iii. variation of parameters.

iv. variation of parameter

v. undetermined co-efficients and variation of parameter

vi. Variation of parameters.